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General Introduction

Modern enterprise networks must ensure reliable connectivity, efficient data transport, and strong security mechanisms. To achieve these objectives, organizations rely on advanced networking technologies such as MPLS, redundant LAN architectures, and monitoring and security solutions.

The objective of this project is to design and implement a complete network infrastructure that integrates these different technologies in a coherent architecture. The project focuses on the creation of a service provider backbone, the deployment of an extended enterprise LAN, and the integration of monitoring and security mechanisms.

The first chapter presents the design and implementation of an MPLS Layer 3 VPN network, which enables secure and isolated communication between multiple customer sites through a shared provider backbone.

The second chapter focuses on the design and implementation of an extended LAN architecture. This part of the project introduces redundancy mechanisms and network design principles to ensure high availability and reliability of the enterprise network.

The third chapter is dedicated to the implementation of monitoring and security solutions. It presents the deployment of a monitoring system and the integration of security services in order to supervise and protect the network infrastructure.

Finally, this report concludes with a general evaluation of the implemented solutions and the results obtained during the different validation tests.

I. Design and Implementation of an MPLS-VPN Network

I.1 Introduction

This chapter presents the design and implementation of the MPLS Layer 3 VPN solution developed in this project. The objective is to understand how a service provider can interconnect multiple customer networks while ensuring traffic separation and efficient routing within the provider backbone.

The chapter first introduces the main technologies used in MPLS VPN environments. It then presents the architecture adopted for the network, the working environment used for the implementation, and the different configuration steps required to deploy the MPLS VPN infrastructure.

Finally, verification procedures and connectivity tests are presented in order to validate the correct operation of the implemented solution.

I.2 MPLS VPN Technology

I.2.1 Multiprotocol Label Switching (MPLS)

Multiprotocol Label Switching (MPLS) is a forwarding technology widely used in service provider networks. Instead of performing a routing table lookup at every hop, MPLS assigns short labels to packets. These labels are then used by routers in the backbone to forward packets quickly and efficiently.

MPLS improves forwarding performance, simplifies packet handling in the core network, and supports advanced services such as traffic engineering and Virtual Private Networks.

I.2.2 Virtual Private Networks (VPN)

A Virtual Private Network (VPN) enables private communication between remote sites over a shared network infrastructure. The main objective of a VPN is to maintain logical separation between different customer networks while allowing each customer to communicate as if using a dedicated private network.

VPN services are widely used by enterprises and service providers because they offer secure interconnection between distant sites and ensure traffic isolation on the shared backbone.

I.2.3 MPLS Layer 3 VPN

MPLS Layer 3 VPN combines MPLS forwarding with routing virtualization mechanisms in order to transport multiple independent customer networks over the same

provider infrastructure.

In this architecture, customer routing information is stored in separate VRF routing tables on the Provider Edge routers. The Provider routers located in the core are responsible for forwarding labeled packets, while the Customer Edge routers connect customer sites to the provider network. Multiprotocol BGP (MP-BGP) is used to distribute VPN routes between PE routers.

This approach allows customer networks to communicate across the MPLS backbone while maintaining strict separation between different VPNs.

I.3 Architecture of the MPLS VPN Solution

The following figure illustrates the global architecture implemented in this project. It shows the service provider backbone, the provider routers, and the different customer sites connected through the MPLS network.

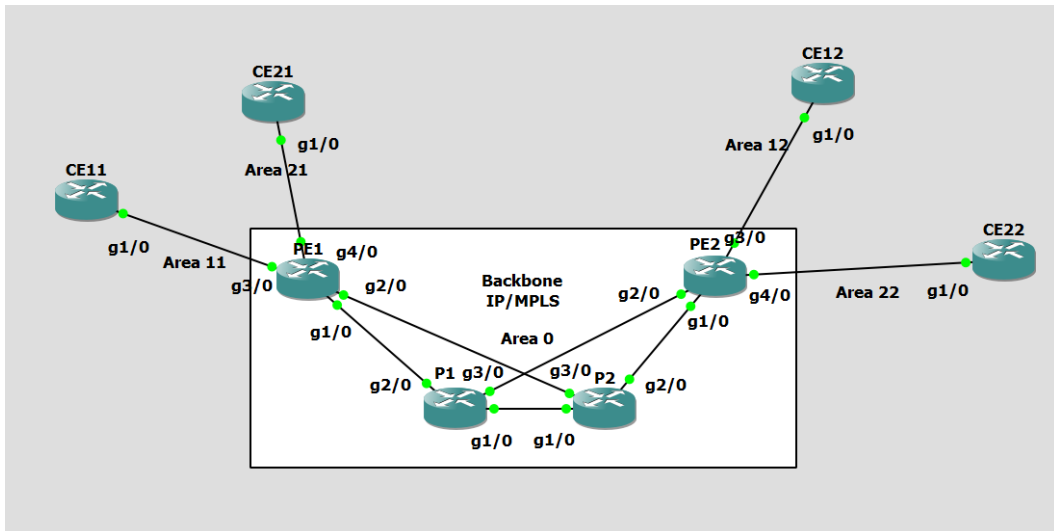


Figure 1: Architecture of the MPLS-VPN solution

The architecture is composed of a provider backbone network and multiple customer networks connected through Provider Edge routers. The backbone is responsible for transporting customer traffic using MPLS while maintaining isolation between different VPNs.

I.3.1 Network Architecture Overview

The provider network forms the MPLS core infrastructure responsible for forwarding labeled packets between different sites. Customer networks are connected to the provider through Customer Edge routers which exchange routing information with the provider edge routers.

This architecture allows multiple customer networks to share the same backbone infrastructure while maintaining logical separation between their routing tables.

1.3.2 Router Roles

Different types of routers are used in the MPLS-VPN architecture, each with a specific role in the network.

Provider Edge (PE) routers are responsible for connecting customer networks to the provider backbone. They maintain VRF routing tables and exchange VPN routing information with other PE routers.

Provider (P) routers are located in the backbone and are responsible only for forwarding labeled packets inside the MPLS core. They do not contain customer routing information.

Customer Edge (CE) routers are located at the customer sites and connect the local networks to the provider infrastructure. These routers exchange routing information with the PE routers.

1.3.3 Routing Architecture

Dynamic routing protocols are used to ensure connectivity between routers and to distribute network routes.

Inside the provider backbone, OSPF is used to establish routing between the provider routers and maintain reachability across the MPLS core.

Between the Provider Edge routers and the Customer Edge routers, OSPF is also used to exchange routing information within each customer network.

1.3.4 VPN Segmentation

Traffic separation between different customers is achieved through the use of Virtual Routing and Forwarding (VRF) instances configured on the Provider Edge routers.

Each customer network is associated with a dedicated VRF routing table. This ensures that routing information from one customer cannot be accessed by another customer, even though they share the same provider infrastructure.

This mechanism guarantees secure and isolated communication between sites belonging to the same customer while preventing communication between different VPNs.

I.4 Working Environment

I.4.1 GNS3 Network Emulator



Figure 2: GNS3 interface used to build and manage the MPLS-VPN topology

The practical implementation of this project was carried out using GNS3 as the main network emulation platform. GNS3 was used to create the MPLS-VPN topology, instantiate the routers, interconnect the devices, and execute the required configurations for the backbone and customer networks.

For the router emulation, a Cisco IOS image for the c7200 platform was imported into GNS3 and used to create the router template. The figures below illustrate the IOS image selection and the resulting router template configuration.

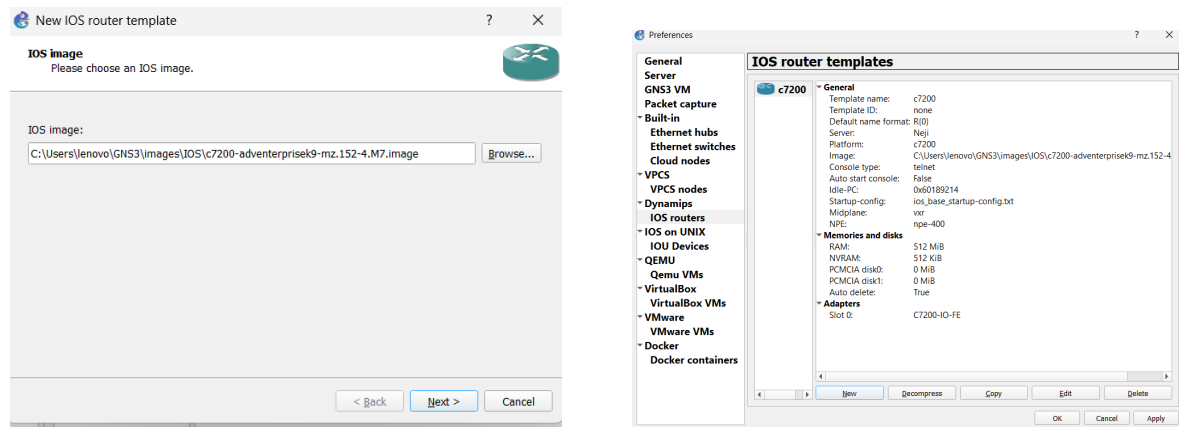


Figure 3: Import of the Cisco IOS image and creation of the c7200 router template

The console access of the routers was configured with the `telnet` console type. This setting was applied in the node properties in order to allow command-line access to the routers during the configuration and verification phases.

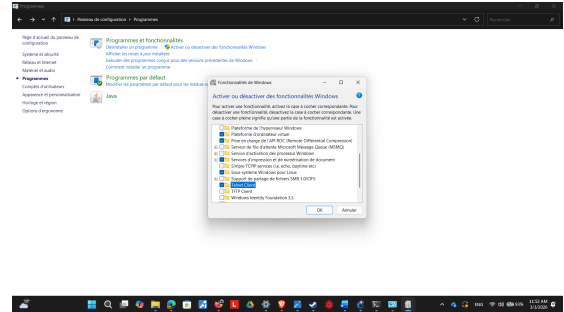
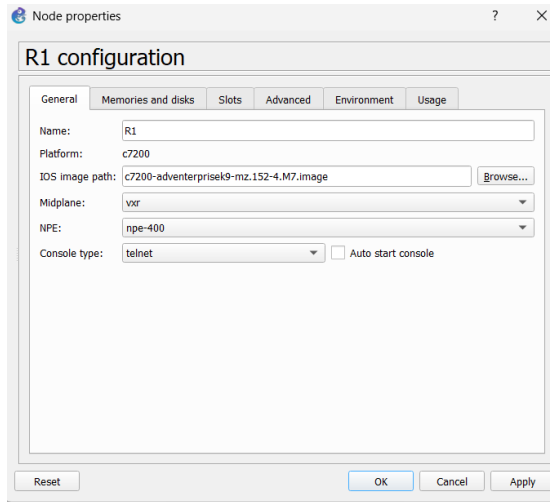


Figure 4: Telnet console configuration in GNS3 and activation of the Telnet Client feature in Windows

In addition, the console command in GNS3 was configured to use Windows Terminal. This configuration allowed direct access to the router command-line interface in a more practical way during the implementation of the project.

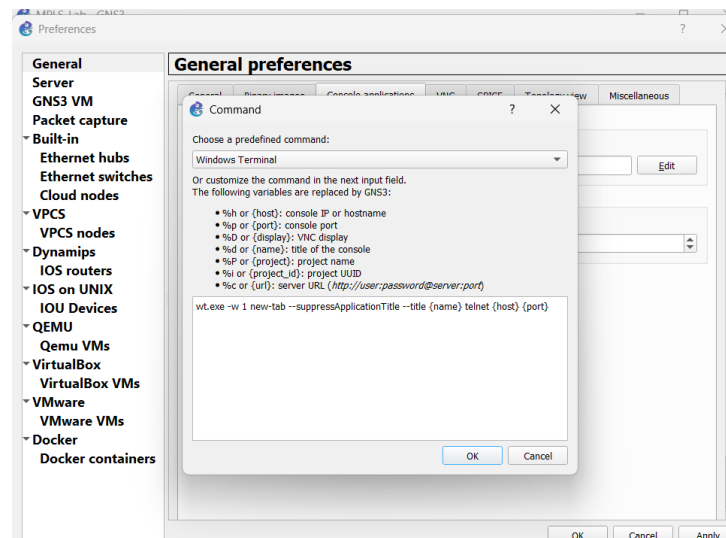


Figure 5: Configuration of Windows Terminal as the console command in GNS3

During the preparation of the laboratory environment, a conflict related to the default UDP tunneling parameters was encountered. The conflict occurred between GNS3 and NVIDIA software running on the host machine. An error appeared when GNS3 attempted to create the UDP NIO, which prevented the topology from running correctly. To solve this issue, the UDP tunneling port range in GNS3 was modified from the default settings to the interval 30000-40000. This adjustment resolved the conflict previously observed on the machine.

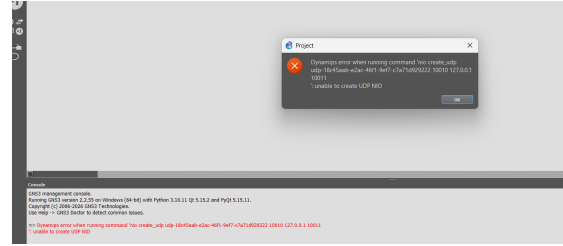
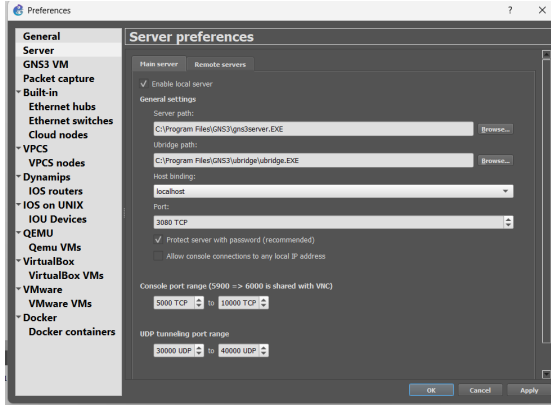


Figure 6: Modification of the UDP tunneling port range and error message related to the UDP conflict in GNS3

I.4.2 VMware Environment

At the current stage of this project, the implementation was completed directly on the local Windows machine using GNS3. No VMware virtual machine was used in the practical realization of this part.

I.4.3 Software and IOS Images Used

The following table summarizes the main software tools and the IOS image used during the implementation of the MPLS-VPN solution.

Tool / Software	Version/Type	Purpose
GNS3	2.2.55	Network emulation and topology implementation
Windows Terminal	1.23.2021.0	Router console access from GNS3
Telnet Client (Windows feature)	Enabled	Command-line access to router consoles
Cisco IOS image	c7200	Router operating system used in GNS3

Table 1: Software and environment used in the project

I.5 Enterprise Network Topology

The following figure presents the enterprise network topology implemented in the GNS3 environment for the MPLS-VPN project.

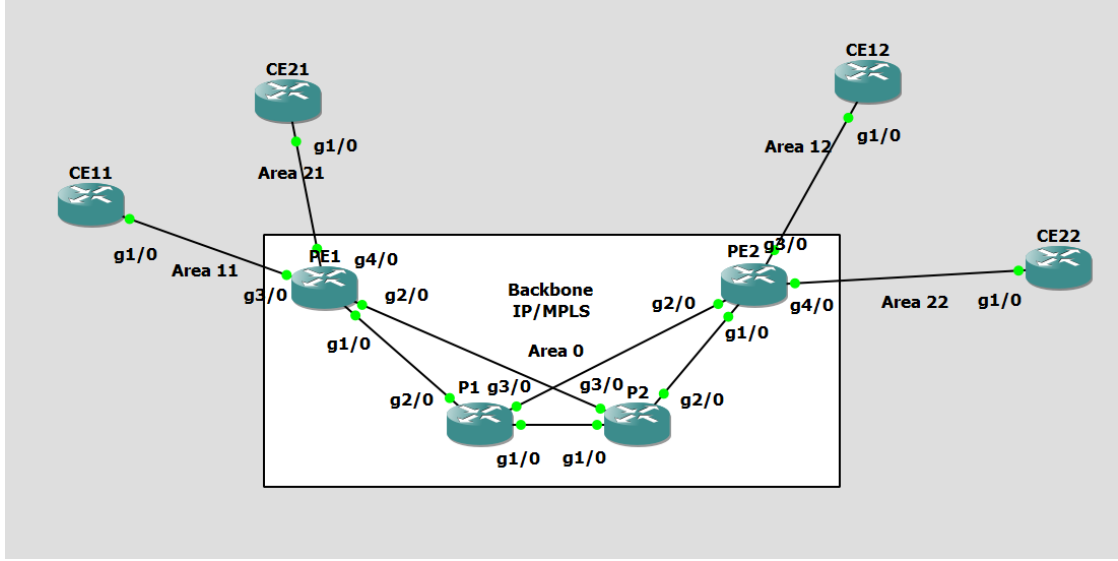


Figure 7: Enterprise network topology implemented in GNS3

The implemented topology is composed of a service provider backbone and two customer networks. The backbone includes two Provider Edge routers, PE1 and PE2, and two Provider routers, P1 and P2. The customer side is represented by four Customer Edge routers: CE11 and CE12 for Customer 1, and CE21 and CE22 for Customer 2.

This topology was designed to simulate a realistic MPLS-VPN environment in which multiple customer sites communicate through a shared provider infrastructure while remaining logically separated.

1.5.1 Router Interface Configuration in GNS3

In order to reproduce the required topology in the GNS3 environment, the router interface modules were configured to provide the necessary number of Gigabit Ethernet interfaces.

Each router was configured with several PA-GE modules in the available slots of the Cisco c7200 platform. This configuration allowed the routers to support multiple connections between the provider backbone and the customer networks.

This step is necessary in GNS3 because the default router template does not always include enough interfaces to build complex network topologies. By adding additional interface modules, the routers can support all the links required in the MPLS backbone and the customer access networks.

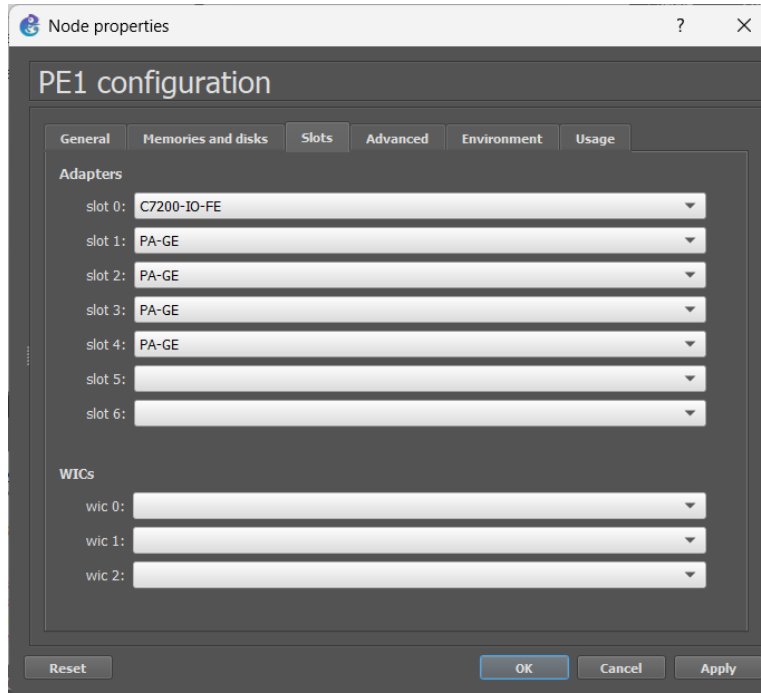


Figure 8: Configuration of interface modules (PA-GE) in the Cisco 7200 router template

1.5.2 Physical and Logical Topology

From a physical point of view, the topology corresponds to the interconnection of the virtual routers created in GNS3. PE1 and PE2 are connected to the provider backbone and also to the customer routers. P1 and P2 form the MPLS core and ensure packet transport inside the provider network.

From a logical point of view, the network is divided into a backbone area and multiple customer areas. The provider backbone belongs to OSPF Area 0, while each customer site is associated with a dedicated OSPF area. In this project, the customer sites are organized as Area 11, Area 12, Area 21, and Area 22.

1.5.3 Backbone IP/MPLS Network

The backbone IP/MPLS network is formed by PE1, PE2, P1, and P2. This part of the topology is responsible for transporting customer traffic across the provider infrastructure using MPLS mechanisms.

The two PE routers act as the entry and exit points of the MPLS cloud for the customer networks, while the P routers operate only inside the provider core. The backbone is configured to ensure internal routing, label distribution, and packet forwarding between the different provider devices.

1.5.4 Customer Networks

Two customer networks are connected to the provider backbone in this topology. Customer 1 is represented by routers CE11 and CE12, while Customer 2 is represented by routers CE21 and CE22.

Each CE router represents a customer site connected to a PE router. Through the MPLS-VPN service, the two sites of the same customer can communicate with each other across the provider backbone. At the same time, communication between different customers remains isolated through the use of separate VRF instances on the PE routers.

I.6 Implementation of the MPLS VPN

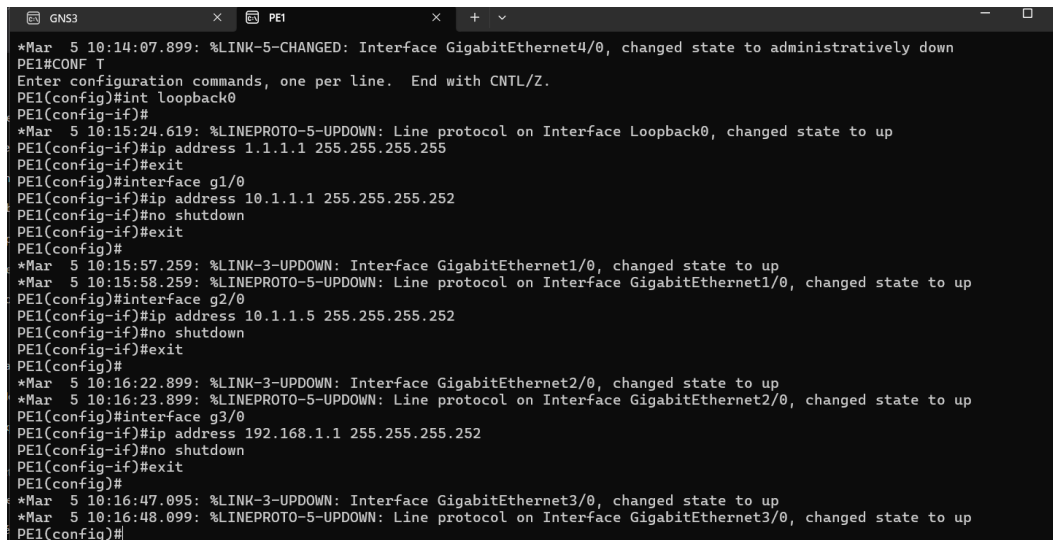
I.6.1 Interface Addressing Configuration

The first step of the implementation consists of configuring the IP addresses on all router interfaces. Each router interface was assigned an address according to the addressing plan defined for the MPLS backbone and the customer networks.

The configuration includes:

- Loopback interfaces used to identify the routers
- Interfaces connecting the provider backbone routers
- Interfaces connecting the Provider Edge routers to the Customer Edge routers

As an example, the following figures present the interface configuration performed on router **PE1**. The loopback interface and the Gigabit Ethernet interfaces were configured with their respective IP addresses and activated using the `no shutdown` command.



```
*Mar 5 10:14:07.899: %LINK-5-CHANGED: Interface GigabitEthernet4/0, changed state to administratively down
PE1#CONF T
Enter configuration commands, one per line. End with CNTL/Z.
PE1(config)#int loopback0
PE1(config-if)#
*Mar 5 10:15:24.619: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
PE1(config-if)#ip address 1.1.1.1 255.255.255.255
PE1(config-if)#exit
PE1(config)#interface g1/0
PE1(config-if)#ip address 10.1.1.1 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#
*Mar 5 10:15:57.259: %LINK-3-UPDOWN: Interface GigabitEthernet1/0, changed state to up
*Mar 5 10:15:58.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0, changed state to up
PE1(config)#interface g2/0
PE1(config-if)#ip address 10.1.1.5 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#
*Mar 5 10:16:22.899: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed state to up
*Mar 5 10:16:23.899: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet2/0, changed state to up
PE1(config)#interface g3/0
PE1(config-if)#ip address 192.168.1.1 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#
*Mar 5 10:16:47.095: %LINK-3-UPDOWN: Interface GigabitEthernet3/0, changed state to up
*Mar 5 10:16:48.099: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet3/0, changed state to up
PE1(config)#
```

Figure 9: PE1 interface configuration (part 1)

```

PE1(config)#interface g4/0
PE1(config-if)#ip address 192.168.1.5 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#
*Mar 5 10:17:18.735: %LINK-3-UPDOWN: Interface GigabitEthernet4/0, changed state to up
*Mar 5 10:17:19.735: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet4/0, changed state to up
PE1(config)#exit
PE1#
*Mar 5 10:17:25.175: %SYS-5-CONFIG_I: Configured from console by console
PE1#copy run start
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
PE1#show ip int brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet1/0 10.1.1.1 YES manual up up
GigabitEthernet2/0 10.1.1.5 YES manual up up
GigabitEthernet3/0 192.168.1.1 YES manual up up
GigabitEthernet4/0 192.168.1.5 YES manual up up
Loopback0 1.1.1.1 YES manual up up
PE1#

```

Figure 10: PE1 interface configuration (part 2) and verification

After configuring the interfaces, the command `show ip interface brief` was used to verify that all interfaces were correctly configured and operational.

The following figures show the interface status for the remaining routers in the topology.

Router	Interface	IP-Address	OK?	Method	Status	Protocol
P1#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	10.1.1.21	YES	manual	up	up
	GigabitEthernet2/0	10.1.1.1	YES	manual	up	up
	GigabitEthernet3/0	10.1.1.10	YES	manual	up	up
	Loopback0	2.2.2.2	YES	manual	up	up
	P1#					
P2#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	10.1.1.22	YES	manual	up	up
	GigabitEthernet2/0	10.1.1.10	YES	manual	up	up
	GigabitEthernet3/0	10.1.1.5	YES	manual	up	up
	Loopback0	0.0.0.0	YES	manual	up	up
	P2#					
P3#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	10.1.1.9	YES	manual	up	up
	GigabitEthernet2/0	10.1.1.13	YES	manual	up	up
	GigabitEthernet3/0	192.168.1.9	YES	manual	up	up
	GigabitEthernet4/0	192.168.1.13	YES	manual	up	up
	P3#					

Figure 11: Interface verification for provider routers

Router	Interface	IP-Address	OK?	Method	Status	Protocol
C11#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	172.16.11.2	YES	manual	up	up
	Loopback0	172.16.11.11	YES	manual	up	up
	C11#					
C12#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	172.16.12.10	YES	manual	up	up
	Loopback0	172.16.12.12	YES	manual	up	up
	C12#					
C22#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	172.16.21.10	YES	manual	up	up
	Loopback0	172.16.21.21	YES	manual	up	up
	C22#					

Figure 12: Interface verification for customer routers

Router	Interface	IP-Address	OK?	Method	Status	Protocol
CE22#	FastEthernet0/0	unassigned	YES	unset	administratively down	down
	GigabitEthernet1/0	192.168.1.14	YES	manual	up	up
	Loopback0	172.16.22.22	YES	manual	up	up
	CE22#					

Figure 13: Interface verification for router CE22

These verification outputs confirm that all router interfaces are active and correctly configured before proceeding with the routing and MPLS configuration steps.

1.6.2 OSPF Configuration in the Backbone

In order to provide routing inside the provider backbone, the OSPF dynamic routing protocol was deployed on the core routers of the MPLS infrastructure.

The backbone operates in **OSPF Area 0**, which represents the central routing domain of the provider network. All provider routers (PE and P) participate in this area in order to exchange routing information and ensure reachability across the MPLS core.

Each router advertises:

- its loopback interface, which uniquely identifies the router
- the backbone interfaces connecting the provider routers

As an example, the following figure presents the OSPF configuration performed on router **P2**. The configuration defines the OSPF process and advertises the relevant networks into Area 0.

```
P2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
P2(config)#router ospf 1
P2(config-router)#network 4.4.4.4 0.0.0.0 area 0
P2(config-router)#network 10.1.1.4 0.0.0.3 area 0
P2(config-router)#network 10.1.1.8 0.0.0.3 area 0
P2(config-router)#
*Mar 5 11:04:21.819: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet3/0 from LOADING to FULL, Loading Done
P2(config-router)#network 10.1.1.20 0.0.0.3 area 0
*Mar 5 11:04:25.719: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet2/0 from LOADING to FULL, Loading Done
P2(config-router)#network 10.1.1.20 0.0.0.3 area 0
P2(config-router)#
*Mar 5 11:04:34.075: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on GigabitEthernet1/0 from LOADING to FULL, Loading Done
P2(config-router)#end
P2#copy ru
*Mar 5 11:04:39.931: %SYS-5-CONFIG_I: Configured from console by console
P2#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
P2#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:35	10.1.1.21	GigabitEthernet1/0
2.2.2.2	1	FULL/DR	00:00:36	10.1.1.9	GigabitEthernet2/0
1.1.1.1	1	FULL/DR	00:00:32	10.1.1.5	GigabitEthernet3/0

```
P2#
```

Figure 14: OSPF configuration and neighbor verification on router P2

After configuring OSPF, the adjacency between routers was verified using the command `show ip ospf neighbor`. The output confirms that router P2 successfully established OSPF neighbor relationships with the other backbone routers.

The routing table was then checked using the command `show ip route ospf`. This verification confirms that the router has learned the routes of the other routers through OSPF.

```

P2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
O       1.1.1.1 [110/2] via 10.1.1.5, 00:01:04, GigabitEthernet3/0
    2.0.0.0/32 is subnetted, 1 subnets
O       2.2.2.2 [110/2] via 10.1.1.9, 00:01:04, GigabitEthernet2/0
    3.0.0.0/32 is subnetted, 1 subnets
O       3.3.3.3 [110/2] via 10.1.1.21, 00:00:54, GigabitEthernet1/0
   10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O       10.1.1.0/30 [110/2] via 10.1.1.21, 00:00:54, GigabitEthernet1/0
           [110/2] via 10.1.1.5, 00:01:04, GigabitEthernet3/0
O       10.1.1.12/30 [110/2] via 10.1.1.21, 00:00:54, GigabitEthernet1/0
           [110/2] via 10.1.1.9, 00:01:04, GigabitEthernet2/0
P2#

```

Figure 15: OSPF routes learned by router P2

These results confirm that the backbone routing infrastructure is correctly established and that all provider routers can exchange routing information through OSPF Area 0.

1.6.3 MPLS Configuration in the Backbone

After establishing IP connectivity through OSPF, MPLS was activated on the provider backbone routers in order to enable label switching inside the network.

MPLS uses labels to forward packets through the provider core instead of relying only on traditional IP routing. This mechanism improves forwarding efficiency and enables advanced services such as MPLS Layer 3 VPN.

The MPLS configuration in the backbone consists of the following steps:

- Enabling Cisco Express Forwarding (CEF)
- Activating the MPLS label distribution protocol (LDP)
- Defining the router identifier using the loopback interface
- Enabling MPLS on the backbone interfaces

The following figures illustrate the MPLS configuration applied to the backbone routers.

```

PE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE1(config)#ip cef
PE1(config)#mpls label protocol ldp
PE1(config)#mpls ldp router-id loopback0 force
PE1(config)#int g1/0
PE1(config-if)#mpls ip
PE1(config-if)#exit
PE1(config)#int g2/0
PE1(config-if)#mpls ip
PE1(config-if)#end
PE1#
*Mar  5 13:59:35.907: %SYS-5-CONFIG_I: Configured from console by console
PE1#

```

Figure 16: MPLS activation on router PE1

```

P1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
P1(config)#ip cef
P1(config)#mpls label protocol ldp
P1(config)#mpls ldp router-id loopback0 force
P1(config)#int g1/0
P1(config-if)#mpls ip
P1(config-if)#exit
P1(config)#int g2/0
P1(config-if)#mpls ip
P1(config-if)#exit
*Mar  5 14:02:22.335: %LDP-5-NBRCHG: LDP Neighbor 1.1.1.1:0 (1) is UP
P1(config-if)#exit
P1(config)#int g3/0
P1(config-if)#mpls ip
P1(config-if)#
*Mar  5 14:02:40.767: %LDP-5-NBRCHG: LDP Neighbor 2.2.2.2:0 (2) is UP
P1(config-if)#end
P1#
*Mar  5 14:02:47.499: %SYS-5-CONFIG_I: Configured from console by console
P1#

```

Figure 17: MPLS activation on router P1

```

P2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
P2(config)#ip cef
P2(config)#mpls label protocol ldp
P2(config)#mpls ldp router-id loopback0 force
P2(config)#int g1/0
P2(config-if)#mpls ip
P2(config-if)#
*Mar 5 14:02:20.803: %LDP-5-NBRCHG: LDP Neighbor 3.3.3.3:0 (1) is UP
P2(config-if)#exit
P2(config)#int g2/0
P2(config-if)#mpls ip
P2(config-if)#
*Mar 5 14:02:37.279: %LDP-5-NBRCHG: LDP Neighbor 2.2.2.2:0 (2) is UP
P2(config-if)#exit
P2(config)#int g3/0
P2(config-if)#mpls ip
P2(config-if)#mpls ip
P2(config-if)#
*Mar 5 14:03:13.119: %LDP-5-NBRCHG: LDP Neighbor 1.1.1.1:0 (3) is UP
P2(config-if)#end
P2#
*Mar 5 14:03:20.091: %SYS-5-CONFIG_I: Configured from console by console
P2#

```

Figure 18: MPLS activation on router P2

Once MPLS is enabled on the backbone interfaces, the Label Distribution Protocol (LDP) automatically establishes neighbor relationships between adjacent routers.

The command `show mpls ldp neighbor` allows verification that LDP sessions have been successfully established between routers.

```

P2#show mpls ldp neighbor
Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 4.4.4.4:0
TCP connection: 3.3.3.3.646 - 4.4.4.4.32693
State: Oper; Msgs sent/rcvd: 13/13; Downstream
Up time: 00:01:23
LDP discovery sources:
GigabitEthernet1/0, Src IP addr: 10.1.1.21
Addresses bound to peer LDP Ident:
10.1.1.21      10.1.1.2      10.1.1.14      3.3.3.3
Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 4.4.4.4:0
TCP connection: 2.2.2.2.646 - 4.4.4.4.58100
State: Oper; Msgs sent/rcvd: 13/15; Downstream
Up time: 00:01:06
LDP discovery sources:
GigabitEthernet2/0, Src IP addr: 10.1.1.9
Addresses bound to peer LDP Ident:
10.1.1.9      10.1.1.13      192.168.1.9      192.168.1.13
2.2.2.2
Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 4.4.4.4:0
TCP connection: 1.1.1.1.646 - 4.4.4.4.18187
State: Oper; Msgs sent/rcvd: 12/14; Downstream
Up time: 00:00:30
LDP discovery sources:
GigabitEthernet3/0, Src IP addr: 10.1.1.5
Addresses bound to peer LDP Ident:
10.1.1.1      10.1.1.5      192.168.1.1      192.168.1.5
1.1.1.1

```

Figure 19: Verification of LDP neighbor relationships

Finally, the MPLS forwarding table was examined using the command `show mpls forwarding-table`. This table shows the labels assigned to different prefixes and the outgoing interfaces used for packet forwarding.

```

P2#
P2#show mpls forwarding-table
Local   Outgoing   Prefix      Bytes Label  Outgoing   Next Hop
Label   Label      or Tunnel Id  Switched     interface
16      Pop Label  1.1.1.1/32   0            Gi3/0       10.1.1.5
17      Pop Label  2.2.2.2/32   0            Gi2/0       10.1.1.9
18      Pop Label  3.3.3.3/32   0            Gi1/0       10.1.1.21
19      Pop Label  10.1.1.0/30   0            Gi3/0       10.1.1.5
        Pop Label  10.1.1.0/30   0            Gi1/0       10.1.1.21
20      Pop Label  10.1.1.12/30  0            Gi2/0       10.1.1.9
        Pop Label  10.1.1.12/30  0            Gi1/0       10.1.1.21
P2#

```

Figure 20: MPLS label forwarding table

The outputs confirm that MPLS is correctly operational in the backbone network and that label switching is functioning between the provider routers.

1.6.4 VRF Configuration on Provider Edge Routers

In MPLS Layer 3 VPN networks, traffic separation between customers is achieved through the use of Virtual Routing and Forwarding (VRF) instances. A VRF creates an independent routing table on the Provider Edge (PE) router, allowing multiple customer networks to coexist on the same infrastructure while remaining logically isolated.

In this project, two VRF instances were created to represent the two customers:

- **VPN_Customer1**
- **VPN_Customer2**

Each VRF is defined using two important parameters:

- **Route Distinguisher (RD)** used to uniquely identify VPN routes in MP-BGP.
- **Route Target (RT)** used to control the import and export of VPN routes between PE routers.

The following figure illustrates the creation of the VRF instances on router **PE1**.

```

PE1(config)#ip vrf VPN_Customer1
PE1(config-vrf)#rd 100:1
PE1(config-vrf)#route-target both 100:1
PE1(config-vrf)#exit
PE1(config)#ip vrf VPN_Customer2
PE1(config-vrf)#rd 100:2
PE1(config-vrf)#route-target both 100:2
PE1(config-vrf)#end
PE1#copy run
*Mar  5 14:26:47.503: %SYS-5-CONFIG_I: Configured from console by console
PE1#copy run start
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
PE1#

```

Figure 21: Creation of VRF instances on router PE1

After creating the VRFs, the customer-facing interfaces must be associated with the corresponding VRF instance using the command `ip vrf forwarding`. This operation places the interface into the appropriate customer routing table.

The following figure shows the assignment of PE1 interfaces to the corresponding VRFs.

```

PE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE1(config)#int g3/0
PE1(config-if)#ip vrf forwarding VPN_Customer1
% Interface GigabitEthernet3/0 IPv4 disabled and address(es) removed due to enabling VRF VPN_Customer1
PE1(config-if)#ip address 192.168.1.1 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#int g4/0
PE1(config-if)#ip vrf forwarding VPN_Customer2
% Interface GigabitEthernet4/0 IPv4 disabled and address(es) removed due to enabling VRF VPN_Customer2
PE1(config-if)#ip address 192.168.1.5 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#end
PE1#
*Mar  5 14:32:22.167: %SYS-5-CONFIG_I: Configured from console by console
PE1#

```

Figure 22: Assignment of PE1 interfaces to VRF instances

The same configuration was then applied to router **PE2**, which connects the remote customer sites to the MPLS backbone.


```

PE2#
PE2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE2(config)#ip vrf VPN_Customer1
PE2(config-vrf)#rd 100:1
PE2(config-vrf)#route-target both 100:1
PE2(config-vrf)#exit
PE2(config)#ip vrf VPN_Customer2
PE2(config-vrf)#rd 100:2
PE2(config-vrf)#route-target both 100:2
PE2(config-vrf)#exit
PE2(config)#interface g3/0
PE2(config-if)#ip vrf forwarding VPN_Customer1
% Interface GigabitEthernet3/0 IPv4 disabled and address(es) removed due to enabling VRF VPN_Customer1
PE2(config-if)#ip address 192.168.1.9 255.255.255.252
PE2(config-if)#no shutdown
PE2(config-if)#exit
PE2(config)#interface g4/0
PE2(config-if)#ip vrf forwarding VPN_Customer2
% Interface GigabitEthernet4/0 IPv4 disabled and address(es) removed due to enabling VRF VPN_Customer2
PE2(config-if)#ip address 192.168.1.13 255.255.255.252
PE2(config-if)#no shutdown
PE2(config-if)#end
PE2#
*Mar  5 14:36:31.939: %SYS-5-CONFIG_I: Configured from console by console
PE2#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
PE2#

```

Figure 23: Creation of VRF instances on router PE2

Through this configuration, each customer network maintains a separate routing table inside the provider edge routers. This mechanism ensures that traffic belonging to one customer cannot interact with the routing information of another customer, thereby guaranteeing secure isolation between VPNs.

1.6.5 OSPF Configuration Between CE and PE

In order to exchange routing information between the Customer Edge (CE) routers and the Provider Edge (PE) routers, the Open Shortest Path First (OSPF) protocol was configured at each customer site.

OSPF is used as the routing protocol inside the customer networks to dynamically advertise the local LAN prefixes toward the provider edge routers. Through the MPLS VPN architecture, these routes are later transported across the provider backbone using MP-BGP.

To maintain logical separation between the different customer sites, each site is configured with a dedicated OSPF area. In this project the following areas were used:

- Area 11 – Customer 1 Site 1
- Area 12 – Customer 1 Site 2
- Area 21 – Customer 2 Site 1
- Area 22 – Customer 2 Site 2

Each CE router advertises two types of networks:

- The link between the CE router and the PE router

- The internal LAN network of the customer site

The following figures show the OSPF configuration performed on the customer edge routers.

```
CE11#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CE11(config)#router ospf 1
CE11(config-router)#network 192.168.1.0 0.0.0.3 area 11
CE11(config-router)#network 172.16.11.11 0.0.0.0 area 11
CE11(config-router)#end
CE11#copy run
```

Figure 24: OSPF configuration on router CE11

```
CE12#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CE12(config)#router ospf 1
CE12(config-router)#network 192.168.1.8 0.0.0.3 area 12
CE12(config-router)#network 172.16.12.12 0.0.0.0 area 12
```

Figure 25: OSPF configuration on router CE12

```
CE21#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CE21(config)#router ospf 1
CE21(config-router)#network 192.168.1.4 0.0.0.3 area 21
CE21(config-router)#network 172.16.21.21 0.0.0.0 area 21
```

Figure 26: OSPF configuration on router CE21

```
CE22#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CE22(config)#router ospf 1
CE22(config-router)#network 192.168.1.12 0.0.0.3 area 22
CE22(config-router)#network 172.16.22.22 0.0.0.0 area 22
```

Figure 27: OSPF configuration on router CE22

Through this configuration, each CE router advertises its local networks to the corresponding PE router. These routes are then placed inside the appropriate VRF instance and later distributed across the MPLS backbone using MP-BGP.

1.6.6 MP-BGP Configuration Between PE Routers

After configuring VRFs and the CE–PE routing exchanges, Multiprotocol BGP (MP-BGP) was deployed between the Provider Edge routers in order to distribute VPN routes across the MPLS backbone.

In MPLS Layer 3 VPN networks, MP-BGP is used to exchange VPNv4 routes between PE routers. Unlike the core routing protocols used inside the provider backbone, MP-BGP is responsible for transporting customer routes associated with their corresponding VRF instances.

In this project, PE1 and PE2 were configured in the same autonomous system, AS 100. The BGP peering was established using the loopback interfaces of the two PE routers in order to provide stable end-to-end communication through the backbone.

The MP-BGP configuration includes the following main operations:

- Definition of the BGP process on the PE routers
- Establishment of the neighbor relationship between PE1 and PE2
- Activation of the `vpnvp4` address family
- Exchange of extended communities using `send-community both`
- Redistribution of OSPF routes from each VRF into MP-BGP

The following figure shows the MP-BGP configuration performed on router **PE1**.

```
PE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE1(config)#router bgp 100
PE1(config-router)#no bgp default ipv4-unicast
PE1(config-router)#neighbor 2.2.2.2 remote-as 100
PE1(config-router)#neighbor 2.2.2.2 update-source loopback0
PE1(config-router)#address-family vpnvp4
PE1(config-router-af)#neighbor 2.2.2.2 activate
PE1(config-router-af)#neighbor 2.2.2.2 send-community both
PE1(config-router-af)#
PE1(config-router-af)#exit-address-family
PE1(config-router)#address-family ipv4 vrf VPN_Customer1
PE1(config-router-af)#redistribute ospf 100 vrf VPN_Customer1
PE1(config-router-af)#exit-address-family
PE1(config-router)#address-family ipv4 vrf VPN_Customer2
PE1(config-router-af)#redistribute ospf 200 vrf VPN_Customer2
PE1(config-router-af)#exit-address-family
PE1(config-router)#end
PE1#
*Mar  5 14:59:14.435: %SYS-5-CONFIG_I: Configured from console by console
PE1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
PE1#
```

Figure 28: MP-BGP configuration on router PE1

The following figure shows the equivalent configuration performed on router **PE2**. The output also confirms that the BGP adjacency with PE1 has been established successfully.

```

PE2#
PE2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE2(config)#router bgp 100
PE2(config-router)#no bgp default ipv4-unicast
PE2(config-router)#neighbor 1.1.1.1 remote-as 100
PE2(config-router)#neighbor 1.1.1.1 update-source loopback0
PE2(config-router)#address-family vpnv4
PE2(config-router-af)#neighbor 1.1.1.1 activate
PE2(config-router-af)#neighbor 1.1.1.1 send-community both
PE2(config-router-af)#
*Mar 5 15:01:06.175: %BGP-5-ADJCHANGE: neighbor 1.1.1.1 Up
PE2(config-router-af)#exit-address-family
PE2(config-router)#address-family ipv4 vrf VPN_Customer1
PE2(config-router-af)#redistribute ospf 100 vrf VPN_Customer1
PE2(config-router-af)#exit-address-family
PE2(config-router)#address-family ipv4 vrf VPN_Customer2
PE2(config-router-af)#redistribute ospf 200 vrf VPN_Customer2
PE2(config-router-af)#exit-address-family
PE2(config-router)#end
PE2#co
*Mar 5 15:02:04.867: %SYS-5-CONFIG_I: Configured from console by console

```

Figure 29: MP-BGP configuration on router PE2

In addition to the MP-BGP configuration, OSPF was configured inside each VRF on the PE routers in order to exchange routes with the CE routers. These OSPF processes were then redistributed into BGP so that customer routes could be transported across the MPLS backbone.

The following figures illustrate the VRF-aware OSPF configuration on PE1 and PE2.

```

PE1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE1(config)#router ospf 100 vrf VPN_Customer1
PE1(config-router)#redistribute bgp 100 subnets
PE1(config-router)#network 192.168.1.0 0.0.0.3 area 11
PE1(config-router)#
*Mar 5 15:05:51.447: %OSPF-5-ADJCHG: Process 100, Nbr 172.16.11.11 on GigabitEthernet3/0 from LOADING to FULL, Loading Done
PE1(config-router)#exit
PE1(config)#router ospf 200 vrf VPN_Customer2
PE1(config-router)#redistribute bgp 100 subnets
PE1(config-router)#network 192.168.1.4 0.0.0.3 area 21
PE1(config-router)#
*Mar 5 15:06:23.991: %OSPF-5-ADJCHG: Process 200, Nbr 172.16.21.21 on GigabitEthernet4/0 from LOADING to FULL, Loading Done

```

Figure 30: OSPF configuration inside VRFs on router PE1

```

PE2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
PE2(config)#router ospf 100 vrf VPN_Customer1
PE2(config-router)#redistribute bgp 100 subnets
PE2(config-router)#network 192.168.1.8 0.0.0.3 area 12
PE2(config-router)#
*Mar 5 15:08:26.211: %OSPF-5-ADJCHG: Process 100, Nbr 172.16.12.12 on GigabitEthernet3/0 from LOADING to FULL, Loading Done
PE2(config-router)#exit
PE2(config)#router ospf 200 vrf VPN_Customer2
PE2(config-router)#redistribute bgp 100 subnets
PE2(config-router)#network 192.168.1.12 0.0.0.3 area 22
PE2(config-router)#end
PE2#
*Mar 5 15:08:55.091: %SYS-5-CONFIG_I: Configured from console by console

```

Figure 31: OSPF configuration inside VRFs on router PE2

I.7 Validation and Connectivity Tests

After completing the configuration of the backbone routing, MPLS, VRFs, OSPF, and MP-BGP, a set of verification and connectivity tests was carried out in order to validate the correct operation of the MPLS-VPN infrastructure.

The objective of this phase is to confirm that:

- MPLS label distribution is correctly established in the provider backbone
- label switching is operational between provider routers
- VPN routes are correctly exchanged between PE routers through MP-BGP
- VRF routing tables contain the expected customer routes
- communication between sites belonging to the same customer is successful
- communication between different customers remains isolated

I.7.1 Verification Commands

The first validation step consists of verifying the MPLS and routing mechanisms inside the provider network.

The command `show mpls ldp neighbor` was executed on router **P2** in order to verify that Label Distribution Protocol (LDP) neighbor relationships were successfully established with the adjacent provider routers.

```
P2#show mpls ldp neighbor
Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 4.4.4.4:0
TCP connection: 3.3.3.3.646 - 4.4.4.4.39177
State: Oper; Msgs sent/rcvd: 15/15; Downstream
Up time: 00:03:13
LDP discovery sources:
  GigabitEthernet1/0, Src IP addr: 10.1.1.21
Addresses bound to peer LDP Ident:
  10.1.1.21      3.3.3.3      10.1.1.2      10.1.1.14
Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 4.4.4.4:0
TCP connection: 2.2.2.2.646 - 4.4.4.4.33886
State: Oper; Msgs sent/rcvd: 15/15; Downstream
Up time: 00:03:13
LDP discovery sources:
  GigabitEthernet2/0, Src IP addr: 10.1.1.9
Addresses bound to peer LDP Ident:
  10.1.1.9      2.2.2.2      10.1.1.13
Peer LDP Ident: 1.1.1.1:0; Local LDP Ident 4.4.4.4:0
TCP connection: 1.1.1.1.646 - 4.4.4.4.21892
State: Oper; Msgs sent/rcvd: 15/15; Downstream
Up time: 00:03:13
LDP discovery sources:
  GigabitEthernet3/0, Src IP addr: 10.1.1.5
Addresses bound to peer LDP Ident:
  10.1.1.1      1.1.1.1      10.1.1.5
```

Figure 32: Verification of MPLS LDP neighbors on router P2

The command `show mpls forwarding-table` was then used on the same router to verify the label forwarding information maintained by MPLS in the backbone.

```
P2#show mpls forwarding-table
Local   Outgoing   Prefix      Bytes Label  Outgoing   Next Hop
Label   Label      or Tunnel Id Switched      interface
19      Pop Label  3.3.3.3/32  0            Gi1/0       10.1.1.21
20      Pop Label  2.2.2.2/32  2026         Gi2/0       10.1.1.9
21      Pop Label  1.1.1.1/32  1382         Gi3/0       10.1.1.5
22      Pop Label  10.1.1.12/30 0            Gi2/0       10.1.1.9
        Pop Label  10.1.1.12/30 0            Gi1/0       10.1.1.21
23      Pop Label  10.1.1.0/30  0            Gi3/0       10.1.1.5
        Pop Label  10.1.1.0/30  0            Gi1/0       10.1.1.21
P2#
```

Figure 33: Verification of the MPLS forwarding table on router P2

To validate the exchange of VPN routes through MP-BGP, the command `show ip bgp vpnv4 vrf VPN_Customer1` was executed on router **PE1**. The output confirms that VPN routes are correctly handled inside the BGP VPNv4 address family.

```
PE1#show ip bgp vpnv4 vrf VPN_Customer1
BGP table version is 13, local router ID is 1.1.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

      Network          Next Hop           Metric LocPrf Weight Path
Route Distinguisher: 100:1 (default for vrf VPN_Customer1)
*> 172.16.11.11/32    192.168.1.2         2           32768 ?
*>i 172.16.12.12/32    2.2.2.2             2          100      0 ?
*> 192.168.1.0/30     0.0.0.0             0           32768 ?
*>i 192.168.1.8/30    2.2.2.2             0          100      0 ?
PE1#
```

Figure 34: Verification of VPNv4 routes on router PE1

The command `show ip route vrf VPN_Customer1` was also executed on router **PE2** in order to verify that the customer routes are correctly installed in the VRF routing table.

```

PE2#show ip route vrf VPN_Customer1

Routing Table: VPN_Customer1
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      172.16.0.0/32 is subnetted, 2 subnets
B       172.16.11.11 [200/2] via 1.1.1.1, 00:03:49
O       172.16.12.12 [110/2] via 192.168.1.10, 00:05:01, GigabitEthernet3/0
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
B       192.168.1.0/30 [200/0] via 1.1.1.1, 00:03:49
C       192.168.1.8/30 is directly connected, GigabitEthernet3/0
L       192.168.1.9/32 is directly connected, GigabitEthernet3/0
PE2#

```

Figure 35: Verification of the VRF routing table on router PE2

These verification commands confirm that the MPLS backbone, the MP-BGP exchange, and the VRF routing separation are all operating correctly.

1.7.2 Connectivity Tests

After verifying the internal operation of the provider network, end-to-end connectivity tests were performed between customer sites.

The first test consisted of sending ICMP packets from router **CE11** to the remote site router **CE12**. Since both routers belong to **Customer 1**, the communication must be successful through the MPLS-VPN infrastructure.

```

CE11#ping 172.16.12.12
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.12.12, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 172/180/200 ms
CE11#

```

Figure 36: Successful connectivity test between CE11 and CE12

The second test was performed between routers **CE21** and **CE22**, which belong to **Customer 2**. The successful result confirms that the second VPN is also correctly operational.

```

CE21#ping 172.16.22.22
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.22.22, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 124/166/180 ms
CE21#

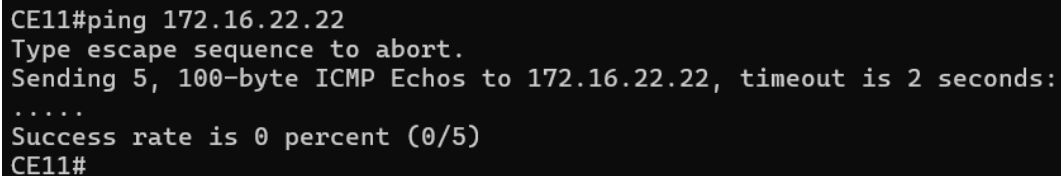
```

Figure 37: Successful connectivity test between CE21 and CE22

1.7.3 VPN Isolation Test

In addition to successful intra-customer communication, it is essential to verify that traffic remains isolated between different customers.

For this purpose, a ping test was performed from router **CE11** toward router **CE21**. Since these two routers belong to different customers and different VRFs, the communication must fail.



```
CE11#ping 172.16.22.22
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.22.22, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
CE11#
```

Figure 38: Isolation test between Customer 1 and Customer 2

The failed ping confirms that the two customer networks remain logically separated, which demonstrates that the MPLS-VPN isolation mechanism is correctly implemented.

Overall, the validation results confirm that the MPLS backbone is functional, VPN routes are correctly distributed, customer sites can communicate within the same VPN, and isolation between different VPNs is preserved.

1.8 Conclusion

This chapter presented the design and implementation of an MPLS-VPN network in a virtual environment. The work included the preparation of the architecture, the configuration of routing and MPLS mechanisms, the creation of VRFs, and the validation of connectivity between customer sites.

The obtained results confirm that MPLS Layer 3 VPN provides a scalable and efficient solution for connecting enterprise networks while ensuring traffic isolation between different customers.

II. Design and Implementation of the Extended LAN Architecture

II.1 Introduction

The second part of the project extends the IP/MPLS infrastructure implemented in Part I by adding an extended enterprise LAN architecture distributed across the headquarters and the two branch sites. While the first part focused mainly on the service-provider backbone, MPLS forwarding, VRF isolation, and inter-site routing between CE routers, this second part focuses on the internal enterprise side of the network, where switching, VLAN segmentation, gateway redundancy, and local access services become essential.

The objective of this chapter is therefore twofold. First, it presents the migration from the multi-VRF customer model studied in Part I to a unified customer domain based on a single `VPN_CYBER` instance. Second, it explains the implementation of the redundant LAN architecture built at the headquarters and the inter-VLAN access architecture implemented at the branches.

This chapter also highlights the use of the GNS3 VM in VMware, which became necessary in this stage of the work in order to support the switching environment required by the topology. Finally, the chapter ends with the validation phase, where the correct behavior of DHCP, HSRP, OSPF, EtherChannel, MPLS/VRF transport, and end-to-end PC connectivity is verified.

II.2 Architecture of the Extended LAN Network

II.2.1 General Evolution of the Project

Compared with Part I, the main evolution introduced in Part II concerns the customer-side architecture. In the first stage of the project, the MPLS backbone was used to transport separate customer VPNs through multiple VRFs. In Part II, the architecture was simplified and reorganized around a single virtual routing domain called `VPN_CYBER`. All CE-facing interfaces on PE1 and PE2 were moved into this VRF, and the CE route exchange was adapted accordingly.

At the same time, the enterprise architecture itself became more realistic. The headquarters was no longer represented only by customer-edge routers, but by a complete LAN design composed of access switches, Layer 3 federator switches, VLAN-based segmentation, and first-hop redundancy. The branches were also expanded with local switching, management VLANs, data VLANs, and DHCP-based host addressing.

II.2.2 Logical Naming of the Sites

In this second stage, the CE routers were logically renamed to better reflect the role of each site in the enterprise network:

- CE11 → **Siege1**
- CE21 → **Siege2**
- CE12 → **Branch1**
- CE22 → **Branch2**

This naming makes the architecture easier to understand because it clearly distinguishes the headquarters routers from the two remote branch routers.

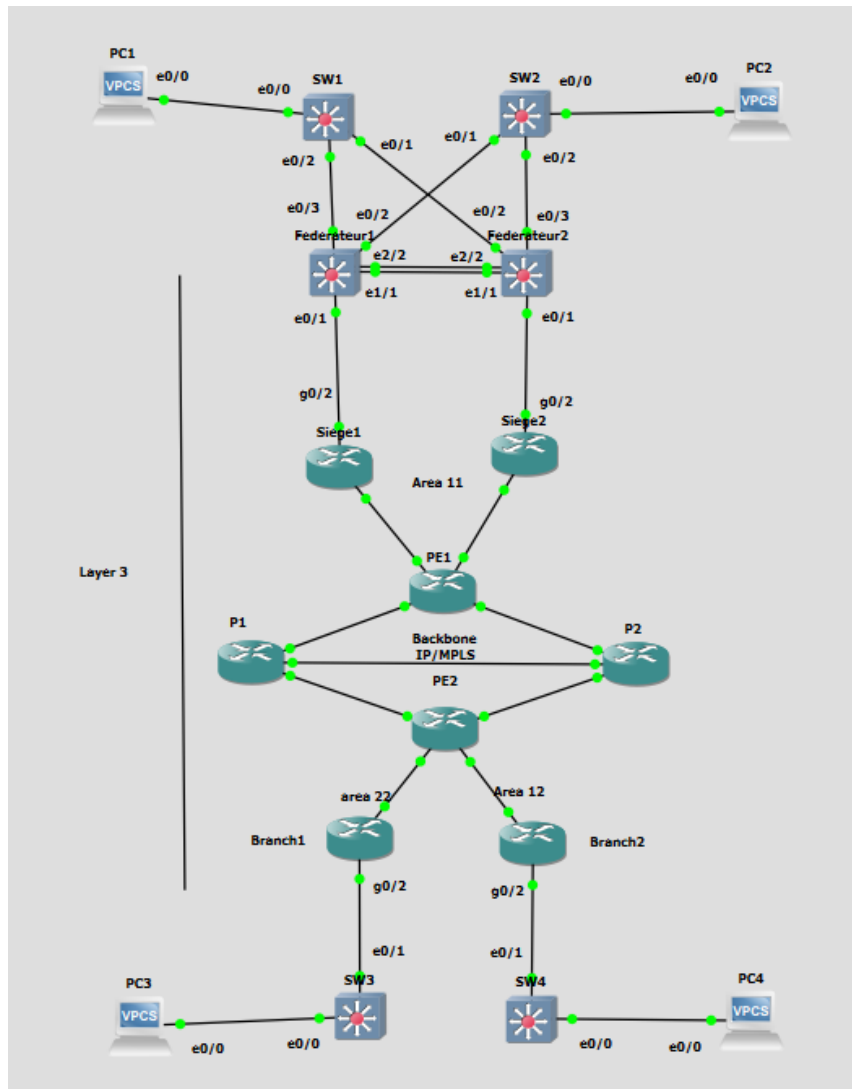


Figure 39: Global topology of Part II. The figure should show the headquarters access layer at the top with SW1 and SW2, the two federator switches in the center, the MPLS/IP backbone in the middle, and the two branch LANs at the bottom. The labels *BackBone IP/MPLS* and *VRF CYBER* should remain visible in the screenshot.

The global topology clearly shows the layered nature of the design. At the headquarters, two access switches connect end devices to two federator switches. These federator switches then connect the headquarters to the MPLS customer-edge routers. At the branch level, each site uses a simpler router-on-a-stick model. The MPLS/IP backbone remains the common transport infrastructure linking all sites together.

II.3 Design of the LAN Solution

II.3.1 Headquarters Design

The headquarters was designed according to a redundant architecture. It is composed of two Layer 2 access switches, denoted **SW1** and **SW2**, and two Layer 3 aggregation devices, denoted **Federateur1** and **Federateur2**. These two federator switches play a central role in the network because they ensure inter-VLAN routing, first-hop redundancy, DHCP service, and dynamic routing toward the headquarters routers.

Three service VLANs were created at the headquarters:

- **VLAN 10**: user subnet 172.16.210.0/24
- **VLAN 15**: user subnet 172.16.215.0/24
- **VLAN 20**: management subnet 172.16.220.0/24

In addition, a transit VLAN was created only between the federator switches:

- **VLAN 300**: transit subnet 192.168.1.44/30

VLAN 10 was used for the first headquarters PC, VLAN 15 for the second headquarters PC, and VLAN 20 for the management interfaces of the headquarters switching devices.

Table 2: Headquarters VLAN plan

VLAN	Purpose	Subnet	Gateway Logic
10	User network 1	172.16.210.0/24	172.16.210.1 is the virtual gateway
15	User network 2	172.16.215.0/24	172.16.215.1 is the virtual gateway
20	Management network	172.16.220.0/24	172.16.220.1 is the virtual gateway
300	Inter-federator transit	192.168.1.44/30	192.168.1.45 and 192.168.1.46

II.3.2 Branch Design

The two branch sites were implemented according to a router-on-a-stick model. Each branch contains one access switch and one CE router. The switch-to-router link is configured as a trunk, while the router uses two subinterfaces to provide inter-VLAN routing.

For **Branch1**, the VLAN structure is the following:

- **VLAN 201**: management subnet 172.16.201.0/24
- **VLAN 202**: data subnet 172.16.202.0/24

For **Branch2**, the VLAN structure is the following:

- **VLAN 203**: management subnet 172.16.203.0/24
- **VLAN 204**: data subnet 172.16.204.0/24

Table 3: Branch VLAN and management plan

Site	VLAN	Role	Addressing
Branch1	201	Management	SW3: 172.16.201.100/24, gateway 172.16.201.1
Branch1	202	Data	Users served by DHCP from Branch1 router
Branch2	203	Management	SW4: 172.16.203.100/24, gateway 172.16.203.1
Branch2	204	Data	Users served by DHCP from Branch2 router

II.4 Description of the Redundant LAN Solution

II.4.1 Migration to VPN_CYBER

Before building the switching part of the enterprise network, the MPLS customer architecture had to be adapted to the new requirement imposed in Part II. A third VRF named **VPN_CYBER** was created on the PE routers. Then, all CE-facing interfaces were moved into this new VRF. Finally, the former customer-specific OSPF processes were replaced by an **OSPF 300** instance bound to **VPN_CYBER**.

This change allowed all four customer routers to exchange routes inside the same MPLS Layer 3 VPN. On the headquarters side, the area assignment was also adjusted by changing the former Area 21 relation into Area 11 for PE1 and Siege2, so that both headquarters routers belonged to one single logical site.

```

Enter configuration commands, one per line.  End with CNTL/Z.
PE1(config)#ip vrf VPN_CYBER
PE1(config-vrf)#rd 100:3
PE1(config-vrf)#route-target both 100:3
PE1(config-vrf)#end
PE1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Mar 20 13:38:07.423: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
Building configuration...
[OK]
PE1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
PE1(config)#int g3/0
PE1(config-if)#no ip add
PE1(config-if)#
*Mar 20 13:39:35.915: %OSPF-5-ADJCHG: Process 100, Nbr 172.16.11.11 on Gigab
itEthernet3/0 from FULL to DOWN, Neighbor Down: Interface down or detached
PE1(config-if)#ip vrf forwarding VPN_CYBER
PE1(config-if)#ip address 192.168.1.1 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#int g4/0
PE1(config-if)#no ip add
PE1(config-if)#
*Mar 20 13:40:20.991: %OSPF-5-ADJCHG: Process 200, Nbr 172.16.21.21 on Gigab
itEthernet4/0 from FULL to DOWN, Neighbor Down: Interface down or detached
PE1(config-if)#ip vrf forwarding VPN_CYBER
PE1(config-if)#ip address 192.168.1.5 255.255.255.252
PE1(config-if)#no shutdown
PE1(config-if)#exit
PE1(config)#end
PE1#wr
Building configuration...
*Mar 20 13:40:54.227: %SYS-5-CONFIG_I: Configured from console by console[OK]

```

Figure 40: Modification of the provider-edge configuration for Part II. The screenshot should show the creation of VPN_CYBER, the migration of CE-facing interfaces into that VRF, and the creation of router ospf 300 vrf VPN_CYBER. A second screenshot may also show the BGP address-family vpnv4 section with redistribution of OSPF 300.

II.4.2 Redundancy Mechanisms at the Headquarters

The headquarters core was intentionally designed with multiple redundancy mechanisms. First, two federator switches were deployed instead of one. Second, the links between the headquarters switching devices were configured as trunks in order to transport the service VLANs across the infrastructure. Third, the two physical links between Federateur1 and Federateur2 were grouped into a single logical EtherChannel in trunk mode. Finally, first-hop redundancy was implemented by means of HSRP on VLANs 10, 15, and 20.

The gateway addressing logic used in the project was the following:

- the first usable address of each VLAN subnet is the HSRP virtual gateway
- the second usable address is assigned to Federateur1
- the third usable address is assigned to Federateur2

Thus, the default gateways seen by the hosts are:

- 172.16.210.1 for VLAN 10
- 172.16.215.1 for VLAN 15
- 172.16.220.1 for VLAN 20

Federateur1 was configured as the main active gateway, while Federateur2 played the backup role. This design ensures that host traffic continues to be forwarded even if one of the two core switches fails.

II.5 Working Environment

II.5.1 Use of GNS3 VM in VMware

The implementation of Part II required an environment capable of supporting both routing and switching devices. For this reason, the GNS3 VM was deployed and used through VMware. This stage represented a practical evolution compared with Part I, where the project focused mainly on routers and MPLS functions.

The use of GNS3 VM was essential for two reasons. First, it allowed the user to integrate the switch-based part of the topology, especially for the headquarters core. Second, it ensured more stable execution of nodes requiring a virtualized environment.

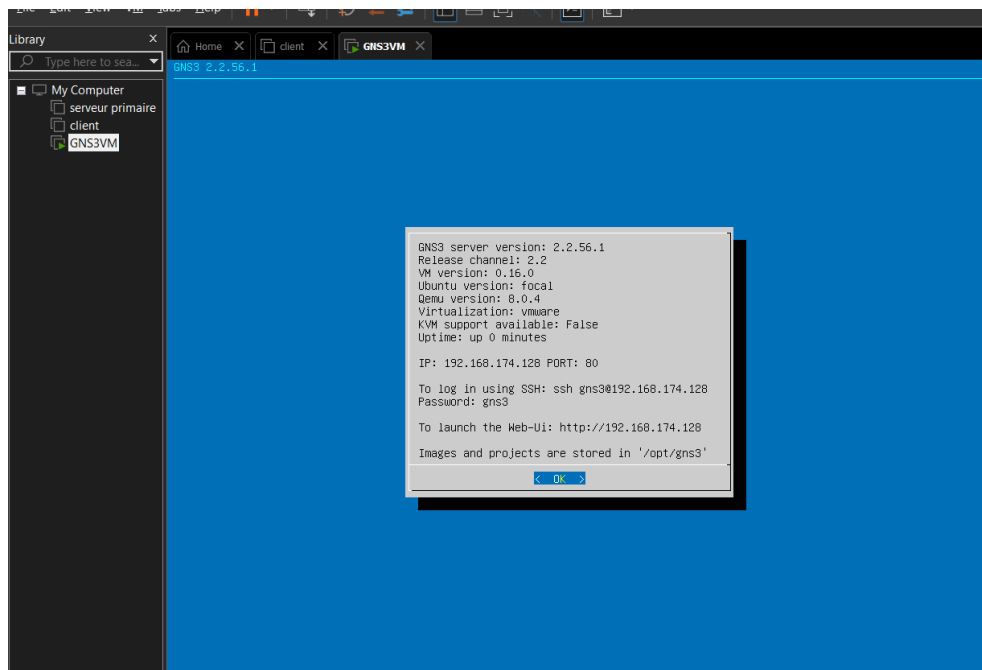


Figure 41: GNS3 VM running in VMware. The image should clearly show that the virtual machine is powered on and connected to the local GNS3 client, since this environment was necessary to support the switching part of Part II.

II.5.2 Logical Devices and Tools

The practical realization of Part II relied on the same router platform used in Part I together with a switching environment supported by the GNS3 VM. The main devices

and tools used in this stage are summarized below.

Table 4: Tools and logical devices used in Part II

Tool / Device	Type	Role in Part II
GNS3	Emulator	Construction and execution of the complete topology
VMware + GNS3 VM	Virtual platform	Support for the switching part of the lab
Cisco routers	Layer 3 devices	MPLS backbone, CE routing, OSPF, VRF, BGP
Layer 2 / Layer 3 switches	Switching devices	VLANs, trunks, HSRP, DHCP, Ether-Channel
Windows Terminal / Telnet	Console access	Direct device configuration and verification

II.6 Implementation of the Redundant LAN Solution

II.6.1 Preparation of the Headquarters Switching Core

The first practical step at the headquarters consisted in preparing the switching devices. The two federator switches were converted into Layer 3 devices by enabling IP routing. Then, VLANs 10, 15, and 20 were created on the four headquarters switches, namely SW1, SW2, Federateur1, and Federateur2.

After the VLAN creation, the user hosts were attached to their proper access VLANs. The PC connected to SW1 was placed in VLAN 10, and the PC connected to SW2 was placed in VLAN 15. The inter-switch links were then configured in trunk mode to carry the VLANs needed by the headquarters architecture.

```

User Access Verification
Username: abdurrahman
Password:
SW1>
SW1#
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#
SW1(config)#vlan 10
SW1(config-vlan)#
SW1(config-vlan)#name ISG1
SW1(config-vlan)#
SW1(config-vlan)#vlan 15
SW1(config-vlan)#
SW1(config-vlan)#name ISG2
SW1(config-vlan)#
SW1(config-vlan)#vlan 20
SW1(config-vlan)#
SW1(config-vlan)#name MANAGEMENT
SW1(config-vlan)#
SW1(config-vlan)#end
SW1#
SW1#wr
Building configuration...
Compressed configuration from 1551 bytes to 978 bytes[OK]
SW1#
*Mar 22 10:59:39.532: %SYS-5-CONFIG_I: Configured from console by abdurrahman on console
SW1#

SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#
SW1(config)#interface range e0/1-2
SW1(config-if-range)#
SW1(config-if-range)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
% Range command terminated because it failed on Ethernet0/1
SW1(config-if-range)#
SW1(config-if-range)#switchport trunk encapsulation dot1q
SW1(config-if-range)#
SW1(config-if-range)#switchport mode trunk
SW1(config-if-range)#
SW1(config-if-range)#switchport trunk native vlan 20
SW1(config-if-range)#
SW1(config-if-range)#
*Mar 22 11:09:43.997: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
*Mar 22 11:09:44.998: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
SW1(config-if-range)#
*Mar 22 11:09:46.004: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
*Mar 22 11:09:47.011: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
SW1(config-if-range)#
*Mar 22 11:09:48.016: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
*Mar 22 11:09:49.020: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
SW1(config-if-range)#
*Mar 22 11:09:50.025: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
*Mar 22 11:09:51.029: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on Ethernet0/1 (20), with Federateur2 Ethernet0/2 (1).
SW1(config-if-range)#

```

Figure 42: creation of vlan and trunk configuration

II.6.2 Management Interfaces, SVIs, HSRP, and DHCP

Once the Layer 2 structure was ready, the management and service interfaces were configured. The management IP addresses of the Layer 2 switches were assigned in VLAN 20 as follows:

- SW1: 172.16.220.101/24
- SW2: 172.16.220.102/24

On the Layer 3 federator switches, the switched virtual interfaces were configured as follows:

- Federateur1: 172.16.210.2, 172.16.215.2, 172.16.220.2
- Federateur2: 172.16.210.3, 172.16.215.3, 172.16.220.3

HSRP was then activated on VLANs 10, 15, and 20 in order to provide default-gateway redundancy. In addition, DHCP pools were configured on both federator switches for the user VLANs. The first ten addresses of each user subnet were excluded, and the remaining addresses were dynamically distributed to hosts. Because both federators act as redundant gateways, both of them were configured with identical DHCP logic.

```
Federateur1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Federateur1(config)#
Federateur1(config)#ip dhcp excluded-address 172.16.210.1 172.16.210.10
Federateur1(config)#
Federateur1(config)#ip dhcp excluded-address 172.16.215.1 172.16.215.10
Federateur1(config)#
Federateur1(config)#ip dhcp pool VLAN10
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#network 172.16.210.0 255.255.255.0
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#default-router 172.16.210.1
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#dns-server 8.8.8.8
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#ip dhcp pool VLAN15
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#network 172.16.215.0 255.255.255.0
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#default-router 172.16.215.1
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#dns-server 8.8.8.8
Federateur1(dhcp-config)#
Federateur1(dhcp-config)#end
Federateur1#
Federateur1#wr
```

Figure 43: SVI, HSRP, and DHCP configuration at the headquarters. The figure should preferably show the SVI IP addresses, the HSRP virtual addresses ending in .1, and the DHCP pool configuration excluding the first ten addresses. A screenshot where Federateur1 appears as the active gateway is ideal.

II.6.3 EtherChannel and VLAN 300 Between the Federators

Because GNS3 does not provide a special dashed cable object for this type of inter-switch redundancy, the two links between Federateur1 and Federateur2 were initially created as two normal physical links. These interfaces were then aggregated using a `channel-group` configuration, resulting in a logical `Port-channel1` operating in trunk mode.

At the same time, VLAN 300 was created only on the two federator switches. This VLAN was used as an inter-federator transit network with the following addresses:

- Federateur1: 192.168.1.45/30
- Federateur2: 192.168.1.46/30

An OSPF cost of 3000 was applied to the VLAN 300 interface in order to preserve the intended routing behavior.

```
Federateur1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Federateur1(config)#
Federateur1(config)#interface vlan 300
Federateur1(config-if)#
*Mar 22 12:49:32.128: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3
00, changed state to down
Federateur1(config-if)#ip address 192.168.1.45 255.255.255.252
Federateur1(config-if)#
Federateur1(config-if)#ip ospf cost 3000
Federateur1(config-if)#
Federateur1(config-if)#no shutdown
Federateur1(config-if)#
Federateur1(config-if)#end
Federateur1#
```

```
Federateur2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Federateur2(config)#
Federateur2(config)#int vlan 300
Federateur2(config-if)#
*Mar 22 12:49:34.563: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3
00, changed state to down
Federateur2(config-if)#ip address 192.168.1.46 255.255.255.252
Federateur2(config-if)#
Federateur2(config-if)#ip ospf cost 3000
Federateur2(config-if)#
Federateur2(config-if)#no shutdown
Federateur2(config-if)#
Federateur2(config-if)#end
Federateur2#
Federateur2#wr
```

Figure 44: therChannel and VLAN 300 configuration between Federateur1 and Federateur2

II.6.4 Routed Uplinks and OSPF Integration at the Headquarters

The links between the federator switches and the headquarters routers were configured as routed links. On the federator side, the corresponding interfaces were converted from switchports to Layer 3 interfaces using the `no switchport` command. On the router side, the interfaces were already Layer 3 by default.

OSPF was then enabled for the headquarters integration using **Area 11**. This point is particularly important because it reflects the logical redesign of the headquarters, where the former Area 21 side was merged into Area 11. The transit VLAN 300 and the routed uplinks therefore participate in the same headquarters routing logic.

The verification stage showed that Federateur1 formed one OSPF adjacency with Siegel on the 192.168.1.28/30 uplink and another adjacency with Federateur2 through VLAN 300. This result confirmed that the headquarters routing core was operating correctly.

```

Siegel1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES NVRAM   administratively down  down
GigabitEthernet1/0     192.168.1.2     YES NVRAM   up                    up
GigabitEthernet2/0     192.168.1.30    YES manual  up                    up
Loopback0              172.16.11.11    YES NVRAM   up                    up
Siegel1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.1.45    1     FULL/DR         00:00:38    192.168.1.29  GigabitEthernet2/0
192.168.1.5     1     FULL/DR         00:00:32    192.168.1.1   GigabitEthernet1/0
Siegel1#

```

Figure 45: OSPF integration at the headquarters. The best screenshot for this section is one showing the routed interface up/up on Federateur1, the OSPF neighbor with Siegel1, and the second neighbor through VLAN 300 toward Federateur2.

II.6.5 Branch1 Implementation

Branch1 was implemented using one access switch and one router. The switch carried VLAN 201 for management and VLAN 202 for user data. The switch-to-router link was configured as a trunk with VLAN 201 as the native VLAN.

On the router side, inter-VLAN routing was implemented with subinterfaces. The management subinterface provided the gateway for VLAN 201, while the data subinterface provided the gateway for VLAN 202. The switch management interface was configured with address 172.16.201.100/24. DHCP was then configured on the Branch1 router for VLAN 202, with the first ten addresses reserved.

```

Branch1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch1(config)#ip dhcp excluded-address 172.16.202.1 172.16.202.10
Branch1(config)#ip dhcp pool VLAN202
Branch1(dhcp-config)#network 172.16.202.0 255.255.255.0
Branch1(dhcp-config)#default-router 172.16.202.1
Branch1(dhcp-config)#default-router 172.16.202.1
Branch1(dhcp-config)#dns-server 8.8.8.8
Branch1(dhcp-config)#end
Branch1#wr

```

Figure 46: Branch1 implementation. The image should ideally show the switch VLAN creation, the trunk toward the router, the subinterfaces on the router, and the DHCP pool for VLAN 202. If possible, add a second screenshot where the PC at Branch1 receives its address dynamically.

II.6.6 Branch2 Implementation

Branch2 followed the same design pattern as Branch1. VLAN 203 was reserved for management and VLAN 204 for user data. The switch-to-router link was configured as a trunk with VLAN 203 as the native VLAN. The Branch2 router then used subinterfaces to provide inter-VLAN routing and DHCP service for VLAN 204.

The management address of SW4 was configured as 172.16.203.100/24, and the Branch2 router distributed user addresses in the 172.16.204.0/24 subnet after excluding the first ten addresses.

```

Branch2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch2(config)#int g2/0
Branch2(config-if)#no shutdown
Branch2(config-if)#
*Mar 22 13:51:17.099: %LINK-3-UPDOWN: Interface GigabitEthernet2/0, changed
state to up
*Mar 22 13:51:18.099: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigabi
tEthernet2/0, changed state to up
Branch2(config-if)#interface g2/0.203
Branch2(config-subif)#encapsulation dot1Q 203 native
Branch2(config-subif)#ip address 172.16.203.1 255.255.255.0
Branch2(config-subif)#interface g2/0.204
Branch2(config-subif)#encapsulation dot1Q 204
Branch2(config-subif)#ip address 172.16.204.1 255.255.255.0
Branch2(config-subif)#end
Branch2#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Mar 22 13:51:58.111: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
Building configuration...
[OK]

```

Figure 47: Branch2 implementation. The figure should show the same type of information as Branch1: VLAN creation, trunking, router subinterfaces, management addressing, and DHCP for VLAN 204.

II.7 Test and Validation of the Redundant LAN Services

II.7.1 Control-Plane Verification

After the full configuration of Part II, several verification steps were carried out. At the MPLS level, the VPN_CYBER design was validated by checking that all four CE routers were able to exchange routes inside the same customer VPN. At the switching level, the VLAN creation, trunk state, EtherChannel state, and SVI status were checked. At the routing level, OSPF neighbor formation was verified on the federators and on the headquarters routers.

The validation also confirmed that DHCP was functioning correctly on the headquarters and on both branches. The PCs received valid IP addresses, subnet masks, default gateways, and DNS information.

```

PC4> show ip

NAME           : PC4[1]
IP/MASK        : 172.16.204.11/24
GATEWAY        : 172.16.204.1
DNS            : 8.8.8.8
DHCP SERVER    : 172.16.204.1
DHCP LEASE     : 86389, 86400/43200/75600
MAC            : 00:50:79:66:68:03
LPORT         : 20030
RHOST:PORT     : 127.0.0.1:20031
MTU            : 1500

PC4>

```

Figure 48: Validation outputs for Part II. A strong figure for this section may combine screenshots of `show etherchannel summary`, `show ip ospf neighbor`, `show standby brief`, and the host `show ip` outputs after DHCP assignment.

II.7.2 End-to-End Connectivity Validation

The final objective of Part II was to restore and validate complete host connectivity across the extended enterprise network. After the troubleshooting phase, the connectivity objective was finally achieved. The corrective work mainly concerned the headquarters side, where the uplink and OSPF behavior had to be stabilized before correct route propagation could be obtained.

Once the routing behavior was corrected, the following results were observed:

- hosts at the headquarters obtained valid addresses through DHCP;
- the local gateways provided by HSRP responded correctly;
- branch hosts also received valid dynamic addresses in their respective data VLANs;
- the headquarters and branch routes were exchanged correctly through OSPF and the MPLS VPN_CYBER transport;
- end-to-end PC connectivity was finally achieved.

This final result is particularly important because it shows that Part II is not limited to a correct control plane, but also achieves the expected data-plane communication between end devices.

```

PC2> ip dhcp
DDORA IP 172.16.215.11/24 GW 172.16.215.1

PC2> ping 172.16.215.1

84 bytes from 172.16.215.1 icmp_seq=1 ttl=255 time=2.127 ms
84 bytes from 172.16.215.1 icmp_seq=2 ttl=255 time=3.389 ms
84 bytes from 172.16.215.1 icmp_seq=3 ttl=255 time=3.826 ms
84 bytes from 172.16.215.1 icmp_seq=4 ttl=255 time=9.487 ms
84 bytes from 172.16.215.1 icmp_seq=5 ttl=255 time=3.553 ms

PC2> ping 172.16.204.11

84 bytes from 172.16.204.11 icmp_seq=1 ttl=58 time=192.187 ms
84 bytes from 172.16.204.11 icmp_seq=2 ttl=58 time=158.755 ms
84 bytes from 172.16.204.11 icmp_seq=3 ttl=58 time=165.095 ms
84 bytes from 172.16.204.11 icmp_seq=4 ttl=58 time=158.544 ms
84 bytes from 172.16.204.11 icmp_seq=5 ttl=58 time=173.098 ms

PC2> |

```

Figure 49: Final end-to-end validation of Part II. The best screenshot here is a set of host or router ping results proving successful inter-site communication, together with host addressing outputs showing that the PCs are in the correct VLANs with the correct gateways.

II.8 Conclusion

This chapter presented the continuation of the MPLS/VPN project through the implementation of an extended and redundant enterprise LAN architecture. The work started with the migration from the multi-VRF model of Part I to a single VPN_CYBER customer domain. It then continued with the deployment of a redundant headquarters core based on VLANs, trunk links, HSRP, DHCP, EtherChannel, and OSPF. Finally, the two branches were implemented using router-on-a-stick inter-VLAN routing and local DHCP services.

From a practical point of view, this part of the project also marked the transition to a more complete virtual lab environment through the use of the GNS3 VM in VMware, which allowed the switching architecture to be integrated into the topology.

The final tests confirmed that the architecture is operational at both the infrastructure and host levels. The MPLS/IP backbone transports the traffic of the VPN_CYBER domain, the headquarters redundancy mechanisms ensure service continuity, the branch LANs are correctly segmented, and end-to-end PC connectivity is achieved across the different sites. This validates the technical objectives of Part II and prepares the project for the security and monitoring work that belongs to the next chapter.

III. Design and Implementation of the Monitoring and Security Solution

III.1 Introduction

The third part of the project is dedicated to the implementation of the monitoring and security functions required for the management plane of the enterprise network. After completing the MPLS/IP backbone in Part I and the extended LAN architecture in Part II, the next objective was to supervise the infrastructure, control administrative access, and validate service quality measurements.

This chapter therefore presents the design and deployment of the monitoring platform, the integration of the Cisco devices into the supervision system, and the implementation of the local security mechanisms used to protect remote administrative access. It also presents the IP SLA configuration used to evaluate latency, jitter, and packet loss between the branch routers and the headquarters routers.

It is important to note that, in the practical scope adopted for this project, the work focused on monitoring through Zabbix, local administrative authentication, VTY access restriction, and IP SLA validation. AAA server integration, logging centralization, and NTP deployment were intentionally left outside the final practical implementation.

III.2 Choice of the Monitoring Solution

The monitoring platform selected for this project is **Zabbix**. This choice was made for several practical and technical reasons.

First, Zabbix is a complete supervision platform that integrates a database, a monitoring server, a web interface, and a flexible monitoring model based on agents and network protocols. Second, it is well adapted to network supervision because it supports SNMP for routers and switches, which corresponds to the requirements of this project. Third, it provides a clear web interface that simplifies verification and demonstration during the final project presentation.

Another important reason behind this choice is that a prebuilt Zabbix image inside GNS3 may lead to instability or compatibility issues. For this reason, the adopted approach was to install Zabbix manually inside an Ubuntu virtual machine running in VMware. This solution provided more control over the environment and made the integration with the GNS3 topology easier to understand and validate.

From a practical point of view, this stage also represented a learning step because Ubuntu had not been used previously in this project. For that reason, the deployment was performed progressively, starting from the creation of the Ubuntu virtual machine,

then the installation of the Zabbix components, and finally the integration of the monitoring server into the management VLAN of the simulated network.

III.3 Functional Model of the Monitoring Solution

The monitoring model implemented in this project is based on the following principle: the Zabbix server collects monitoring information from the network devices through SNMP, stores the collected values in its database, and displays them through the web interface.

The logical data flow is therefore the following:

- the Cisco devices expose management information through SNMP;
- the Zabbix server communicates with these devices by using their management IP addresses;
- the collected data is stored in the Zabbix database;
- the administrator visualizes the status, values, and alerts through the web interface.

The implementation was centered on the management network. The monitoring server was assigned the address 172.16.220.250/24, while the devices were monitored through their management IP addresses. The ten monitored devices are the following:

Table 5: Devices integrated into Zabbix

Logical name	Management interface	Management IP
Federateur1	VLAN 20	172.16.220.2
Federateur2	VLAN 20	172.16.220.3
SW1	VLAN 20	172.16.220.101
SW2	VLAN 20	172.16.220.102
SW3	VLAN 201	172.16.201.100
SW4	VLAN 203	172.16.203.100
Siege1	Loopback0	172.16.11.11
Siege2	Loopback0	172.16.21.21
Branch1	Loopback0	172.16.12.12
Branch2	Loopback0	172.16.22.22

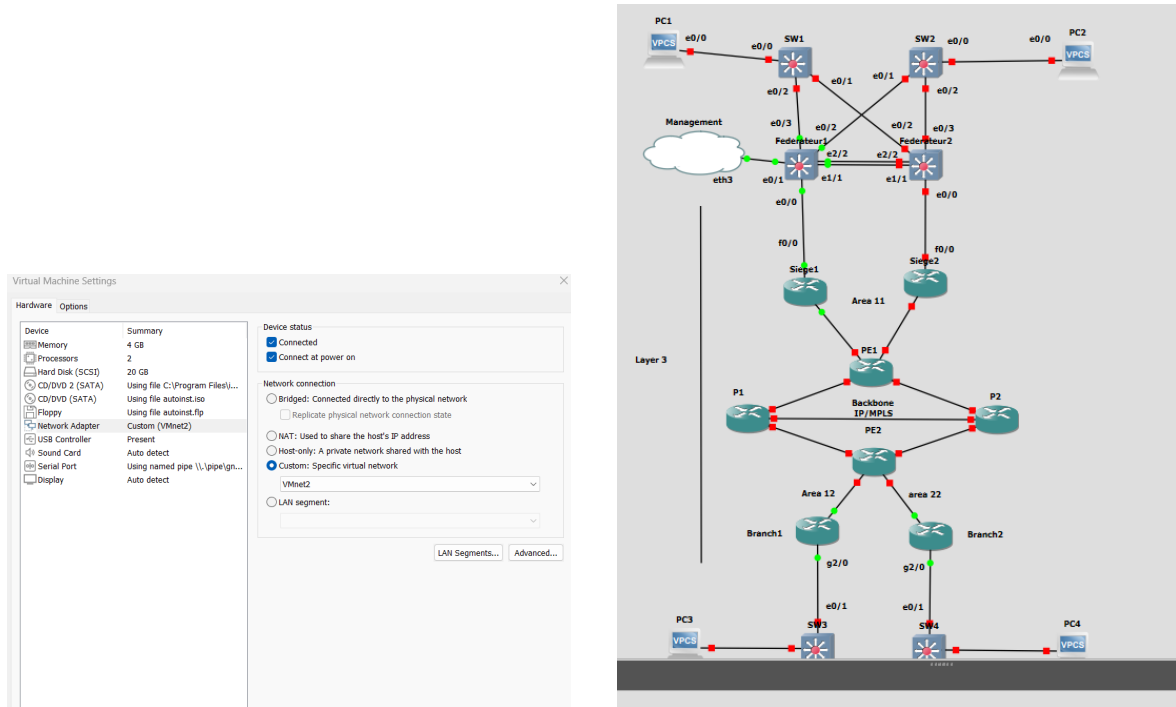


Figure 50: Preparation of the Ubuntu monitoring server and its integration into the GNS3 topology. On the left, the Ubuntu virtual machine is connected to the VMware virtual network VMnet2. On the right, the GNS3 Cloud node is linked to Federateur1 in order to insert the monitoring server into the management network.

```

--- 172.16.220.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3064ms
pipe 3
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ ping -c 4 172.16.220.2
PING 172.16.220.2 (172.16.220.2) 56(84) bytes of data.
64 bytes from 172.16.220.2: icmp_seq=1 ttl=255 time=1.99 ms
64 bytes from 172.16.220.2: icmp_seq=2 ttl=255 time=1.76 ms
64 bytes from 172.16.220.2: icmp_seq=3 ttl=255 time=2.38 ms
64 bytes from 172.16.220.2: icmp_seq=4 ttl=255 time=1.98 ms

--- 172.16.220.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.755/2.024/2.381/0.225 ms
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ ping -c 4 172.16.220.1
PING 172.16.220.1 (172.16.220.1) 56(84) bytes of data.
64 bytes from 172.16.220.1: icmp_seq=1 ttl=255 time=3.31 ms
64 bytes from 172.16.220.1: icmp_seq=2 ttl=255 time=13.1 ms
64 bytes from 172.16.220.1: icmp_seq=3 ttl=255 time=7.37 ms
64 bytes from 172.16.220.1: icmp_seq=4 ttl=255 time=6.37 ms

--- 172.16.220.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 3.305/7.545/13.136/3.558 ms
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$

```

Figure 51: Validation of the Ubuntu server connectivity to the management network through successful ping tests toward the management gateway and Federateur1.

The previous figures illustrate the practical mechanism used to insert the Ubuntu Zabbix server into the simulated enterprise network. First, the Ubuntu virtual machine was attached to the VMware virtual network VMnet2, which was also used as the interconnection basis for the GNS3 environment. Second, the GNS3 Cloud node was connected to **Federateur1** through a dedicated management link in VLAN 20, which

allowed the Ubuntu machine to enter the management plane of the topology. Finally, the successful ping tests toward 172.16.220.1 and 172.16.220.2 confirmed that the monitoring server had effectively become part of the management network. This integration step was essential, because Zabbix could not communicate with the Cisco devices through SNMP before the Ubuntu server obtained valid Layer 3 reachability inside the management subnet.

III.4 Implementation of the Monitoring Solution

III.4.1 Installation of Zabbix on Ubuntu

The monitoring platform was installed on a dedicated Ubuntu virtual machine in VMware. The installation included the three main components required by Zabbix:

- the database;
- the Zabbix server;
- the web interface.

Once the platform was installed, the Ubuntu server was configured with a static IP address in the management network:

- IP address: 172.16.220.250/24
- Default gateway: 172.16.220.1

This step was followed by connectivity validation between the Ubuntu server and the management gateway on the headquarters core.

III.4.2 SNMPv3 Configuration on the Devices

Although the project statement allows SNMP v1, v2c, or v3, the final implementation adopted **SNMPv3** in order to provide stronger authentication. The common SNMPv3 model used on the devices is based on:

- security group: ISG
- user: ISG
- authentication protocol: MD5
- authentication password: SSIR
- Zabbix destination host: 172.16.220.250

However, the exact SNMP configuration was not strictly identical on all device categories. The main difference concerned the **trap-source** and the selected trap families.

Federator switches. For `Federateur1` and `Federateur2`, the SNMP trap source was configured on `Vlan20` because these devices are managed through the headquarters management VLAN. In addition to link traps, HSRP and OSPF-related traps were activated because these two devices are directly involved in the redundant core and the dynamic routing behavior.

```
Federateur1(config)#no snmp-server community SSIR
Federateur1(config)#
Federateur1(config)#snmp-server group ISG v3 auth
Federateur1(config)#
Federateur1(config)#snmp-server user ISG ISG v3 auth md5 SSIR
Federateur1(config)#
Federateur1(config)#
*Apr 21 13:31:07.204: Configuring snmpv3 USM user, persisting snmpEngineBoots. Please Wait...
Federateur1(config)#snmp-server trap-source Vlan20
Federateur1(config)#
Federateur1(config)#snmp-server enable traps snmp linkdown linkup
Federateur1(config)#
Federateur1(config)#snmp-server enable traps hsrp
Federateur1(config)#
Federateur1(config)#$aps ospf cisco-specific state-change nssa-trans-change
Federateur1(config)#
Federateur1(config)#snmp-server enable traps ipsla
Federateur1(config)#
Federateur1(config)#snmp-server host 172.16.220.250 version 3 auth ISG
Federateur1(config)#
Federateur1(config)#
```

Figure 52: SNMPv3 configuration applied to the federator switches

Headquarters and branch routers. For the routers `Siegel1`, `Siegel2`, `Branch1`, and `Branch2`, the trap source was configured on `Loopback0` because the project uses the loopback addresses as the management IPs of these devices. OSPF-related traps were activated on the routers, while `Branch1` and `Branch2` also received IP SLA related traps because these two routers generate the SLA measurements.

```
Siegel1#ANSIO,SI
Siegel1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Siegel1(config)#snmp-server group ISG v3 auth
Siegel1(config)#snmp-server user ISG ISG v3 auth md5 SSIR
Siegel1(config)#snmp-server trap-source Loopback0
Siegel1(config)#snmp-server enable traps snmp linkdown linkup
Siegel1(config)#$aps ospf cisco-specific state-change nssa-trans-change
Siegel1(config)#snmp-server host 172.16.220.250 version 3 auth ISG
Siegel1(config)#end
Siegel1#wr
*Apr 21 15:09:12.731: %SYS-5-CONFIG_I: Configured from console by abdurrahmen on console
Siegel1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
```

Figure 53: SNMPv3 configuration on the headquarters routers

```

Branch1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch1(config)#snmp-server group ISG v3 auth
Branch1(config)#snmp-server user ISG ISG v3 auth md5 SSIR
Branch1(config)#snmp-server trap-source Loopback0
Branch1(config)#snmp-server enable traps snmp linkdown linkup
Branch1(config)#$aps ospf cisco-specific state-change nssa-trans-change
Branch1(config)#snmp-server enable traps ipsla
Branch1(config)#snmp-server host 172.16.220.250 version 3 auth ISG
Branch1(config)#end
Branch1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Apr 21 14:56:44.231: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
Building configuration...
[OK]
Branch1#

```

Figure 54: SNMPv3 configuration on the branch routers

Access switches. For the access switches, the SNMP trap source was bound to the management SVI. Thus, SW1 and SW2 use Vlan20, SW3 uses Vlan201, and SW4 uses Vlan203. In their case, the main activated notifications are related to interface state changes.

```

SW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW2(config)#
SW2(config)#snmp-server group ISG v3 auth
SW2(config)#
SW2(config)#snmp-server user ISG ISG v3 auth md5 SSIR
SW2(config)#
SW2(config)#
*Apr 21 13:47:25.676: Configuring snmpv3 USM user, persisting snmpEngineBoots. Please Wait...

SW2(config)#snmp-server trap-source Vlan20
SW2(config)#
SW2(config)#snmp-server enable traps snmp linkdown linkup
SW2(config)#
SW2(config)#snmp-server host 172.16.220.250 version 3 auth ISG
SW2(config)#
SW2(config)#end
SW2#
SW2#wr
Building configuration...
Compressed configuration from 2165 bytes to 1332 bytes[OK]
SW2#
SW2#
*Apr 21 13:47:47.473: %SYS-5-CONFIG_I: Configured from console by abdurrahmen on console
SW2#

```

Figure 55: SNMPv3 configuration on the access switches

III.4.3 Integration of the Devices in Zabbix

After the Cisco side was prepared, the devices were added one by one in Zabbix through the web interface. The **Generic by SNMP** template was used for all hosts, since the supervision objective was to validate generic network monitoring through SNMP rather than to rely on highly specialized vendor templates.

The same SNMPv3 parameters were reused for all hosts at the Zabbix interface level:

- SNMP version: SNMPv3

- security name: ISG
- security level: authNoPriv
- authentication protocol: MD5
- authentication passphrase: SSIR

Only the host name and the management IP address changed from one device to another.

Host

Host IPMI Tags Macros Inventory Encryption Value mapping

* Host name Federateur1

Visible name Federateur1

Templates Name Actions
Generic by SNMP Unlink Unlink and clear

type here to search Select

* Host groups ISG X type here to search Select

Interfaces	Type	IP address	DNS name	Connect to	Port	Default
SNMP	172.16.220.2		IP	DNS	161	☒ Remove

Add

Description

Update Clone Delete Cancel

Figure 56: General host creation page in Zabbix

Host

Host IPMI Tags Macros Inventory Encryption Value mapping

Interfaces	Type	IP address	DNS name	Connect to	Port	Default
SNMP	172.16.220.2		IP	DNS	161	☒ Remove

* SNMP version SNMPv3

Max repetition count 10

Context name

Security name ISG

Security level authNoPriv

Authentication protocol MD5

Authentication passphrase SSIR

☒ Use combined requests

Add

Description

Update

Figure 57: SNMPv3 parameters used in the Zabbix host configuration

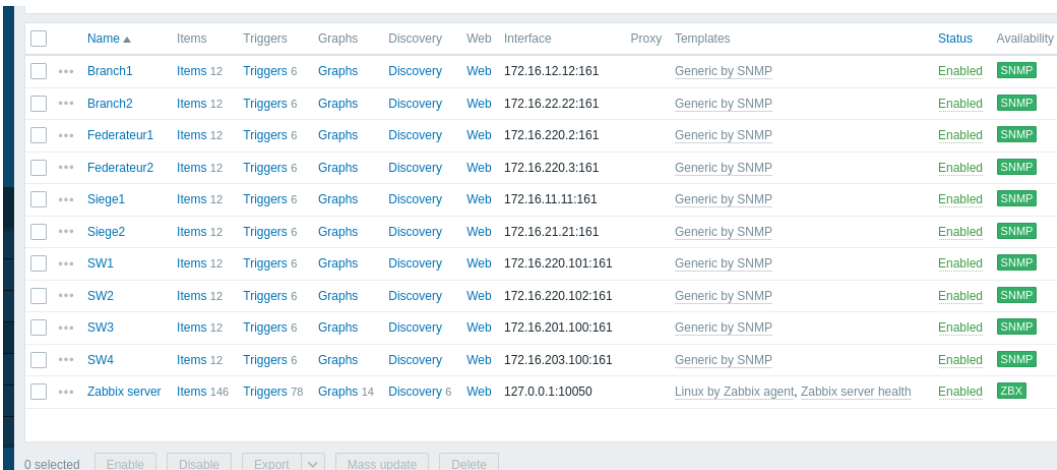
III.5 Test and Validation of the Monitoring Solution

The validation phase of the monitoring solution focused on two complementary points. First, it was necessary to verify basic network reachability between the Ubuntu server and the management interfaces of the devices. Second, it was necessary to verify that

Zabbix was actually receiving SNMP data and displaying monitored values through the web interface.

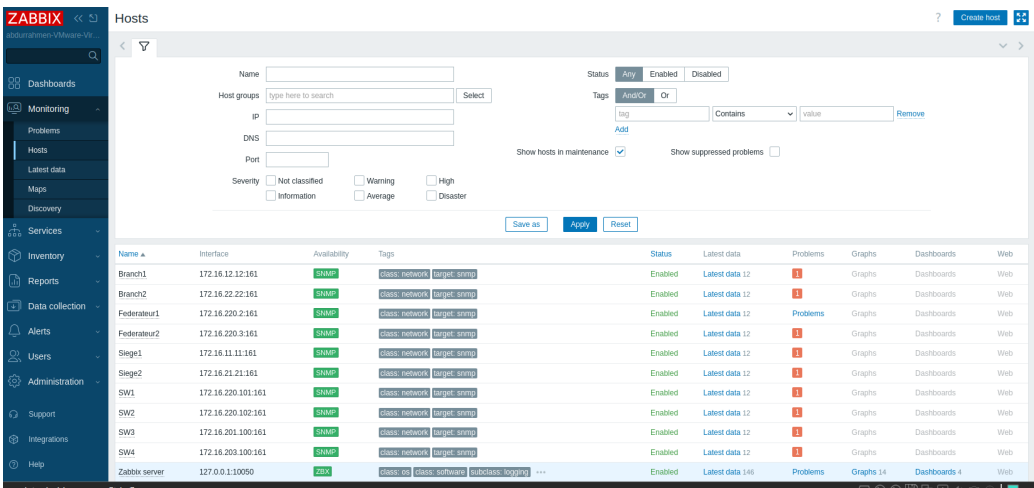
The first successful validation step was obtained with **Federateur1**, which became reachable from the Ubuntu monitoring server. This confirmed that the Ubuntu virtual machine was correctly integrated into VLAN 20 and that the SNMP communication path was operational.

The same methodology was then extended to the rest of the managed devices. After the host entries were created in Zabbix, the availability of the hosts and the collection of values were verified in the Zabbix interface.



☐	Name ▲	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability
☐	*** Branch1	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.12.12:161		Generic by SNMP	Enabled	SNMP
☐	*** Branch2	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.22.22:161		Generic by SNMP	Enabled	SNMP
☐	*** Federateur1	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.220.2:161		Generic by SNMP	Enabled	SNMP
☐	*** Federateur2	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.220.3:161		Generic by SNMP	Enabled	SNMP
☐	*** Siege1	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.11.11:161		Generic by SNMP	Enabled	SNMP
☐	*** Siege2	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.21.21:161		Generic by SNMP	Enabled	SNMP
☐	*** SW1	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.220.101:161		Generic by SNMP	Enabled	SNMP
☐	*** SW2	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.220.102:161		Generic by SNMP	Enabled	SNMP
☐	*** SW3	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.201.100:161		Generic by SNMP	Enabled	SNMP
☐	*** SW4	Items 12	Triggers 6	Graphs	Discovery	Web	172.16.203.100:161		Generic by SNMP	Enabled	SNMP
☐	*** Zabbix server	Items 146	Triggers 78	Graphs 14	Discovery 6	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX

Figure 58: Zabbix host list showing the monitored devices, their IP addresses, and their SNMP availability



Name ▲	Interface	Availability	Tags	Status	Latest data	Problems	Graphs	Dashboards	Web
Branch1	172.16.12.12:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
Branch2	172.16.22.22:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
Federateur1	172.16.220.2:161	SNMP	class: network target: snmp	Enabled	Latest data 12	Problems	Graphs	Dashboards	Web
Federateur2	172.16.220.3:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
Siege1	172.16.11.11:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
Siege2	172.16.21.21:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
SW1	172.16.220.101:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
SW2	172.16.220.102:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
SW3	172.16.201.100:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
SW4	172.16.203.100:161	SNMP	class: network target: snmp	Enabled	Latest data 12	1	Graphs	Dashboards	Web
Zabbix server	127.0.0.1:10050	ZBX	class: os class: software subclass: logging	Enabled	Latest data 146	Problems	Graphs 14	Dashboards 4	Web

Figure 59: Monitoring overview in Zabbix after the integration of the network devices

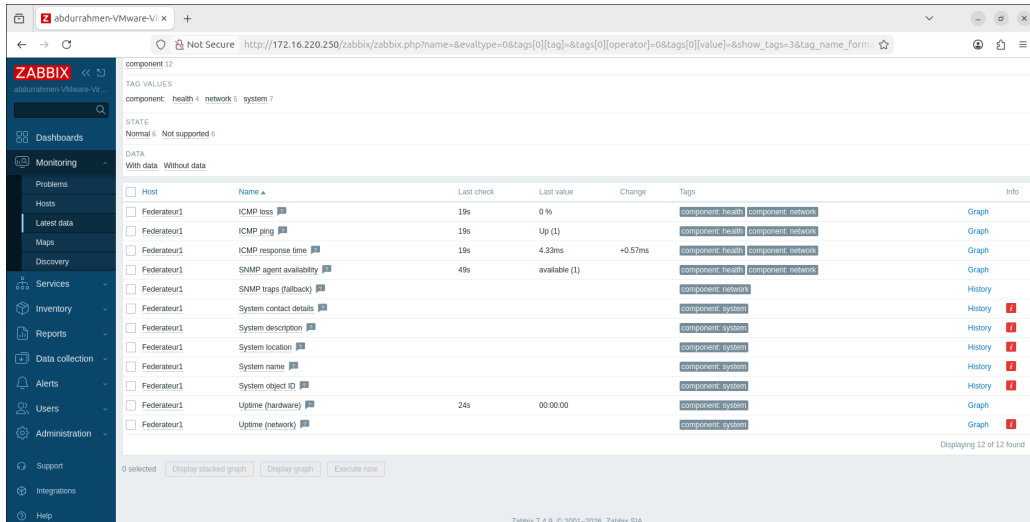


Figure 60: Example of monitored values in Zabbix using Federateur1 as reference

The previous figures confirm that the monitoring server not only stores the host definitions, but also collects operational data from the supervised devices. This validates the practical deployment of the monitoring solution.

III.6 Description of the Security Solution

The security solution implemented in this chapter does not rely on an external AAA server. Instead, it focuses on local device hardening and administrative access restriction directly on the Cisco equipment.

Two security mechanisms were retained:

- local privileged authentication through a locally defined administrative account;
- restriction of remote VTY access to the two authorized administrator IP addresses only.

According to the project statement, the only authorized administrative stations are:

- ADMIN1 = 172.16.220.100/24
- ADMIN2 = 172.16.220.150/24

All other sources must be denied at the VTY level. This logic was implemented by combining `login local`, a local privilege 15 account, and an access control list applied inbound on the VTY lines.

III.7 Working Environment for the Security and Monitoring Phase

The practical environment used in this phase is based on the following elements:

- a Windows host machine running the GNS3 client;
- VMware hosting the GNS3 VM and the Ubuntu monitoring server;
- GNS3 used to build and run the enterprise topology;
- an Ubuntu virtual machine hosting the Zabbix platform;
- Cisco routers and switches emulated in GNS3.

Unlike a classical enterprise directory-based environment, no Active Directory or Windows Server based AAA infrastructure was deployed in the final scope of this work. The administrative security logic was instead implemented directly on the Cisco devices through local credentials and VTY access filtering.

III.8 Implementation of the Security Solution

III.8.1 Local Administrative Authentication

The first security measure consisted of defining a local administrative account with privilege level 15 on the devices. This allowed administrators to access privileged commands immediately after successful authentication, without requiring an additional enable password phase during the validation scenario.

III.8.2 VTY Restriction by ACL

The second security measure consisted of creating a standard ACL and applying it to the VTY lines. The implemented ACL authorizes only the two management stations 172.16.220.100 and 172.16.220.150, and explicitly logs the final deny statement.

This ACL logic was applied in the same way on all the managed devices.

```

Siegel>en
Password:
Siegel#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Siegel(config)#username abdurrahen privilege 15 secret neji
Siegel(config)#ip access-list standard ADMIN_VTY
Siegel(config-std-nacl)#permit 172.16.220.100
Siegel(config-std-nacl)#permit 172.16.220.150
Siegel(config-std-nacl)#deny any log
Siegel(config-std-nacl)#exit
Siegel(config)#line con 0
Siegel(config-line)#login local
Siegel(config-line)#exit
Siegel(config)#line vty 0 4
Siegel(config-line)#login local
Siegel(config-line)#transport input telnet
Siegel(config-line)#access-class ADMIN_VTY in
Siegel(config-line)#end
Siegel#wr
*Apr 22 10:10:32.559: %SYS-5-CONFIG_I: Configured from console by abdurrahe
n on console
Siegel#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]

```

Figure 61: Common local authentication and VTY ACL logic applied on the managed devices

Because the Cisco images used in the laboratory environment did not consistently support SSH on all devices, Telnet was retained as the practical transport for the VTY validation. The important security point in this project is therefore not the transport protocol itself, but the fact that remote administrative access is restricted to the two authorized management IP addresses only.

III.9 Test and Validation of the Security Services

III.9.1 Validation of the VTY Restriction

The first validation point consisted of proving that unauthorized administrative access is blocked. For example, the monitoring server at 172.16.220.250 was not supposed to access the devices through VTY, while ADMIN1 and ADMIN2 had to be accepted.

During the practical validation, the devices correctly exposed their VTY service only to authorized admin addresses. The ACL logic and the local login mechanism therefore behaved as expected.

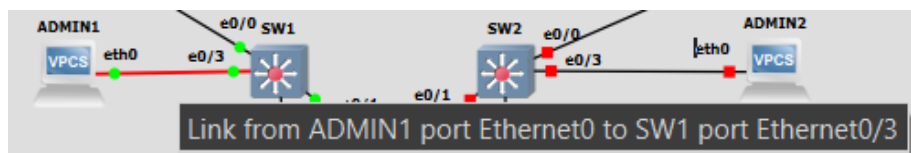


Figure 62: Integration of the administrative stations ADMIN1 and ADMIN2 into the management VLAN through SW1 and SW2.


```

ADMIN1> ip 172.16.220.100/24 172.16.220.1
Checking for duplicate address...
ADMIN1 : 172.16.220.100 255.255.255.0 gateway 172.16.220.1

ADMIN1> save
Saving startup configuration to startup.vpc
. done

ADMIN1> ping 172.16.220.1

84 bytes from 172.16.220.1 icmp_seq=1 ttl=255 time=12.815 ms
84 bytes from 172.16.220.1 icmp_seq=2 ttl=255 time=2.242 ms
84 bytes from 172.16.220.1 icmp_seq=3 ttl=255 time=2.965 ms
84 bytes from 172.16.220.1 icmp_seq=4 ttl=255 time=2.351 ms
84 bytes from 172.16.220.1 icmp_seq=5 ttl=255 time=2.587 ms

ADMIN1> ping 172.16.220.2

84 bytes from 172.16.220.2 icmp_seq=1 ttl=255 time=1.971 ms
84 bytes from 172.16.220.2 icmp_seq=2 ttl=255 time=2.530 ms
84 bytes from 172.16.220.2 icmp_seq=3 ttl=255 time=5.273 ms
84 bytes from 172.16.220.2 icmp_seq=4 ttl=255 time=3.333 ms
84 bytes from 172.16.220.2 icmp_seq=5 ttl=255 time=3.760 ms

ADMIN1>

```

Figure 63: Configuration of ADMIN1 with its management IP address and validation of connectivity toward the management gateway and Federateur1.

```

abdurrahmen@abdurrahmen-VMware-Virtual-Platform: ~
[sudo] password for abdurrahmen:
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ nc -vz -s 172.16.220.100 172.16.220.2 23
nc: connect to 172.16.220.2 port 23 (tcp) failed: Connection timed out
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ nc -vz -s 172.16.220.100 172.16.220.2 23
nc: connect to 172.16.220.2 port 23 (tcp) failed: Connection timed out
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ sudo ip addr del 172.16.220.100/24 dev ens33
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ sudo ip addr del 172.16.220.250/24 dev ens33
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ sudo ip addr add 172.16.220.100/24 dev ens33
abdurrahmen@abdurrahmen-VMware-Virtual-Platform:~$ telnet 172.16.220.2
Trying 172.16.220.2...
Connected to 172.16.220.2.
Escape character is '^]'.

User Access Verification

Username: abdurrahmen
Password:
Federateur1#

```

Figure 64: Validation of remote VTY access from an authorized administrative address through the successful display of the login prompt on Federateur1.

The previous figures illustrate the practical validation of the administrative access policy implemented in Part III. First, the administrative stations ADMIN1 and ADMIN2 were connected to the management VLAN through the headquarters access switches. Then, ADMIN1 was configured with the address 172.16.220.100/24 and successfully

reached both the virtual management gateway 172.16.220.1 and `Federateur1` at 172.16.220.2. For the remote access validation, the VPCS environment was sufficient to confirm IP-level connectivity, but it did not provide a clear interactive demonstration of the VTY login process. For this reason, the final access test was performed from the terminal using an authorized administrative source address, which made it possible to clearly display the `User Access Verification` prompt on the target device. This result confirms that remote access was effectively permitted from an authorized administrative source. Overall, these validations demonstrate the coherence between the Layer 2 integration of the admin stations, the IP addressing plan, and the VTY access restriction logic configured on the Cisco devices.

III.9.2 Validation of the IP SLA Measurements

The final validation step concerned IP SLA. According to the project requirements, the two branch routers had to generate `udp-jitter` tests toward the headquarters routers, while the latter had to operate as responders.

The implemented SLA configuration was based on two operations per branch router:

- operation 11 toward 192.168.1.2
- operation 21 toward 192.168.1.6

A UDP destination port was added in the practical configuration because the IOS image required a complete `udp-jitter` syntax.

```
Branch2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Branch2(config)#ip sla 11
Branch2(config-ip-sla)#udp-jitter 192.168.1.2 5000
Branch2(config-ip-sla-jitter)#frequency 300
Branch2(config-ip-sla-jitter)#precision milliseconds
Branch2(config-ip-sla-jitter)#exit
Branch2(config)#ip sla 21
Branch2(config-ip-sla)#udp-jitter 192.168.1.6 5000
Branch2(config-ip-sla-jitter)#frequency 300
Branch2(config-ip-sla-jitter)#precision milliseconds
Branch2(config-ip-sla-jitter)#exit
Branch2(config)#ip sla schedule 11 start-time now life forever
Branch2(config)#ip sla schedule 21 start-time now life forever
Branch2(config)#end
Branch2#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Apr 22 10:29:36.587: %SYS-5-CONFIG_I: Configured from console by abdurrahme
n on console
[confirm]
```

Figure 65: IP SLA configuration on the branch routers

On the headquarters routers, the `ip sla responder` command was activated in order to answer the branch measurements.

```

Siegel#conf t
Enter configuration commands, one
Siegel(config)#ip sla responder
Siegel(config)#end
Siegel#wr

```

Figure 66: IP SLA responder configuration on the headquarters routers

Once routing convergence was complete, the command `show ip sla statistics` returned successful results, making it possible to observe the status of the operations and the quality indicators associated with them.

```

Branch1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 11
Type of operation: udp-jitter
  Latest RTT: 104 milliseconds
Latest operation start time: 12:01:52 UTC Wed Apr 22 2026
Latest operation return code: OK
RTT Values:
  Number Of RTT: 10                RTT Min/Avg/Max: 84/104/121 millisecon
ds
Latency one-way time:
  Number of Latency one-way Samples: 0
  Source to Destination Latency one way Min/Avg/Max: 0/0/0 millisecond
s
  Destination to Source Latency one way Min/Avg/Max: 0/0/0 millisecond
s
Jitter Time:
  Number of SD Jitter Samples: 9
  Number of DS Jitter Samples: 9
  Source to Destination Jitter Min/Avg/Max: 0/10/24 milliseconds
  Destination to Source Jitter Min/Avg/Max: 0/8/16 milliseconds
Packet Loss Values:
  Loss Source to Destination: 0
  Source to Destination Loss Periods Number: 0
  Source to Destination Loss Period Length Min/Max: 0/0
  Source to Destination Inter Loss Period Length Min/Max: 0/0
  Loss Destination to Source: 0
  Destination to Source Loss Periods Number: 0
  Destination to Source Loss Period Length Min/Max: 0/0
  Destination to Source Inter Loss Period Length Min/Max: 0/0
  Out Of Sequence: 0      Tail Drop: 0
  Packet Late Arrival: 0  Packet Skipped: 0
Voice Score Values:
  Calculated Planning Impairment Factor (ICPIF): 0
  Mean Opinion Score (MOS): 0
Number of successes: 2
Number of failures: 1
Operation time to live: Forever

```

Figure 67: Example of IP SLA statistics on a branch router

The previous figure confirms that the branch routers are effectively measuring the path quality toward the headquarters side. This satisfies the practical objective of validating latency, jitter, and packet-loss related mechanisms.

III.10 Conclusion

This chapter presented the design and implementation of the monitoring and security services associated with the enterprise management plane. The work started with the deployment of a Zabbix supervision server on Ubuntu in VMware and continued with the integration of the monitoring server into the GNS3 environment through the management VLAN.

The ten devices required by the project were then integrated into Zabbix through SNMPv3, using their management addresses and a unified monitoring profile. This phase validated the operational supervision of the infrastructure.

From the security point of view, the practical implementation focused on local administrative authentication and VTY access restriction. Only the two authorized management stations were allowed to reach the remote administrative service, in accordance with the project statement.

Finally, the IP SLA configuration completed the monitoring dimension of the chapter by allowing the branch routers to measure service quality toward the headquarters routers. The successful validation of these mechanisms confirms that the objectives of Part III were achieved from both the monitoring and security perspectives.

General Conclusion

This project made it possible to design, implement, and validate a complete enterprise network architecture combining transport, local network services, supervision, and security mechanisms in a coherent practical environment.

In the first part, the work focused on the implementation of an MPLS Layer 3 VPN infrastructure. This stage made it possible to understand the functioning of the provider backbone, the role of PE, P, and CE routers, and the mechanisms used to guarantee route isolation between customers. The configuration of OSPF, MPLS, VRFs, and MP-BGP confirmed that a shared backbone can transport several logically separated customer networks while preserving secure communication between the sites belonging to the same VPN.

In the second part, the project evolved toward a more realistic enterprise network architecture through the implementation of an extended LAN solution. This stage introduced VLAN segmentation, inter-VLAN routing, HSRP redundancy, DHCP services, EtherChannel, and the integration of the switching environment through the GNS3 VM in VMware. The obtained results confirmed the stability of the headquarters design, the correct operation of the branch architectures, and the restoration of end-to-end communication between the different sites of the enterprise network.

In the third part, the project was extended to the management and supervision plane. A Zabbix monitoring server was deployed on Ubuntu in VMware and integrated into the enterprise management VLAN. The network devices were then added to the supervision platform through their management IP addresses using SNMPv3. In parallel, local administrative security was implemented by combining privilege level 15 local accounts with VTY access restriction based on access control lists. Finally, IP SLA measurements were configured in order to evaluate latency, jitter, and packet loss between the branch routers and the headquarters routers. These results demonstrated that the implemented solution is not limited to connectivity only, but also provides visibility, control, and validation tools for the operation of the network.

From a practical point of view, this project also represented an important learning experience. It required the use of several complementary tools and environments, including GNS3, GNS3 VM, VMware, Ubuntu, Cisco IOS devices, and the Zabbix web interface. Beyond the technical configurations themselves, the project made it possible to develop a better understanding of troubleshooting, topology integration, protocol interaction, and service validation in a virtual enterprise network.

Overall, the obtained results confirm that the proposed architecture satisfies the main objectives of the project. The MPLS backbone ensures inter-site transport, the

extended LAN design provides structured and redundant local connectivity, and the monitoring and security mechanisms strengthen the management of the infrastructure. The final implementation therefore constitutes a complete and coherent network solution that combines performance, availability, supervision, and controlled administrative access.

As a future perspective, this work could be extended by integrating a complete AAA server, centralized logging, and NTP synchronization in order to further strengthen the security and management dimensions of the infrastructure. Additional improvements could also include more advanced Zabbix dashboards, automated alerting policies, and deeper performance analysis